



Review

A survey of UAV-based data collection: Challenges, solutions and future perspectives

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ABSTRACT

Internet of Things (IoT) generates unlimited data, which should be collected and forwarded towards a central controller (CC) for further processing and decision-making. Nevertheless, the unique features of IoT devices, such as their limited energy capacity, large-scale deployment, and prospective mobility, introduce new challenges and constraints, resulting in unoptimized data gathering. Therefore, as a forthcoming solution, Unmanned Aerial Vehicles (UAVs) could be dispatched as auxiliary devices, and the benefits of their potential are fully exploited, such as movement flexibility, mitigation of IoT communication burden, minimization of data collection (DC) delay, powering IoT devices, and efficient delivery of collected data. The current state-of-the-art has a noticeable gap that hinders the systematic reviews of UAV-based DC schemes. To bridge this gap, this survey provides an extensive analysis of UAV-based DC schemes to light up the topic. We first outline the main characteristics of UAV networks, which have to be considered when designing an efficient UAV-based DC scheme. Also, we highlight the main challenges that, if considered during the data gathering process, can significantly enhance the assistance of UAVs to IoT devices. We then describe various UAV-based DC applications with their key features. Next, our paper comprehensively discusses UAV-based DC schemes according to a proposed taxonomy. Finally, we discuss current problems, difficulties, and potential future directions for UAV-based DC research.

1. Introduction

The new generations of mobile networks such as fifth-generation (5G), beyond 5G (B5G), or even sixth-generation (6G) have recently attracted growing research efforts (Guo et al., 2021). Indeed, these networks are expected to connect more than 100 billion Internet of Things (IoT) devices and support up to 1000 times more traffic volume (Zhong et al., 2020c). Also, these connected IoT devices could generate enormous data streams that should be transmitted to a central controller (CC) in order to be processed, leading to the establishment of the Big Data concept (Awaysheh et al., 2021). The efficient processing of this generated data sometimes requires advanced artificial intelligence (AI) methods to efficiently analyze and address the complexity of data (Sasaki, 2021; Amanullah et al., 2020). However, before making

this kind of processing, an efficient data collection (DC) has to be initially carried out while considering the nature of data and the specific features of the IoT devices (Stoyanova et al., 2020). Moreover, when there is a massive deployment of IoT devices, the DC requires dispatching a terrestrial mobile data collector to travel near IoT devices, thus minimizing their energy consumption for data transmission, avoiding interference, and reducing data losses (Liu et al., 2020a). In the last decade, scientists have demonstrated that adopting Unmanned Aerial Vehicles (UAVs) as aerial data collectors could provide superior performances compared to other kinds of data collectors (Huang et al., 2021b). Indeed, the UAVs' mobility is not restricted by the movements of the terrestrial IoT devices on the ground and can be efficiently adapted to IoT devices deployed on a large scale over a complex

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geographical area. Nevertheless, many issues surrounding the assistance of UAVs to IoT devices remain to be examined.

1.1. Motivation

Data collection has recently drawn significant attention in various fields, such as traffic state reporting (Kumar et al., 2021), environmental monitoring (Lin et al., 2020), and physical phenomena measurements (Xu et al., 2020). Each of these applications imposes IoT devices to cover the area of interest and be placed in the right places to acquire the data correctly (Yu et al., 2021; Salameh et al., 2021). Moreover, IoT devices may sense different types of data (e.g., light, temperature, or signals) and need to transmit the sensed data to a CC for processing and analysis (Yazdinejad et al., 2020). However, depending on the distance separating IoT devices from the CC and the mode of communication, this transmission could cause unbalanced energy consumption over IoT devices and data losses, significantly deteriorating the network performance if not addressed correctly (Ding et al., 2020). To address these issues, a ground mobile data collector could be deployed to dynamically move between IoT devices to act as an interface or a relay station with the CC and ensure that all generated data are collected (Zhang and Li, 2020). This strategy can effectively link the remote IoT devices to the CC, address their limited energy constraint, and reduce the data losses caused by interference. Nevertheless, the paths of such data collectors are often restricted by obstacles, which do not allow them to optimally cover and serve the maximum possible IoT devices promptly, mainly when time-sensitive data are generated (Balasubramanian et al., 2021). Additionally, ground mobile data collectors are less effective with dynamic IoT devices than static ones because they are not supposed to adapt to their mobility (Zhu and Wang, 2021).

Recently, UAVs have been widely adopted to assist terrestrial networks, users, and different communicating equipment, allowing terrestrial devices to extend their capabilities and thus establishing an integrated system (Azzoug and Boukra, 2022). Furthermore, an incredible effort has been made in the literature to deploy UAVs as mobile data collectors to both fairly serve IoT devices and optimally be adapted to their unpredictable mobility (Pogaku et al., 2022; Basharat et al., 2022). Therefore, UAVs may provide many advantages to overcome the issues discussed above, and they can be used as creative approaches to address the many difficulties in the next generations of mobile networks (Masaracchia et al., 2021). However, various problems related to deploying UAVs are also distinguished, which are not thoroughly studied and addressed. Indeed, UAVs have a restricted-energy capacity and cannot remain continuously active during the whole flight mission. Also, UAVs face severe constraints related to IoT devices, such as the restricted energy capacity of such devices, their movements, and the nature of their generated data (Shah et al., 2021).

To overcome these challenges, each existing UAV-based DC solution in the literature has been dedicated to tackling a given problem separately. For instance, some solutions are based on classical optimization and AI techniques to optimize the trajectories of UAVs, the generation of data on IoT devices, and the maximization of the collected data amount. Another category of solutions has considered the restricted energy capacity of IoT devices and deploying UAVs to serve as energy transmitters and data collectors for gathering data from IoT devices and powering them. Other UAV-based solutions have also considered time-sensitive data and their associated security aspects. As a result, UAVs have demonstrated their usefulness in servicing IoT devices and their utility in optimizing DC (Duong et al., 2022). Our survey realized that the concept of UAV-based DC has been widely studied in the past six years, and different contributions have been proposed exponentially in the three recent years. Moreover, the motivations behind this study are threefold. First, according to the state-of-the-art, no published survey jointly considers the existing solutions, pros and cons, and various features of UAV-based DC. Second, we distinguished that different emergent technologies are continuously involved in this paradigm,

thus generating, without interruption, new UAV-based DC categories. Finally, we want to propose a survey that comprehensively covers the topic of UAV-based DC and provides scientists and readers with a complete vision of its research directions and future perspectives.

1.2. Comparison with existing surveys

With the increasing interest in UAVs and deploying them to assist terrestrial networks, it comes as no surprise that in the last few years, the topic of UAV-based DC has generated a vast number of solutions. Moreover, some survey articles have also been published, but they did not fully cover the topic or, in most cases, they only superficially discuss specific schemes without categorizing them or accurately mentioning their pros and cons. This sub-section briefly discusses several relevant studies that have been suggested in the last five years. Firstly, some of them discussed energy-efficient DC schemes using UAVs. Jawhar et al. (2017) covered several networking layer protocols used in UAV-based systems and discussed some energy-efficient UAV-based DC over Wireless Sensor Networks (WSNs). Sekander et al. (2018) investigated the disadvantages of conventional single-tier UAV networks over multi-tier UAV network design in terms of energy consumption of UAVs, interference, and spectral efficiency. Finally, Sonawane and Sujithra (2018) mainly focused on the energy efficiency maximization in UAV networks and how UAVs are exploited in localizing and recognition of UAV nodes.

Secondly, some surveys focused on integrating the security and Wireless Power Transfer (WPT) concepts into the UAV-based DC process. Given the restricted energy capacity of ground nodes and UAVs, Pakrooh and Bohlooli (2021) presented a survey on services based on UAV-assisted IoT environments. The different discussed schemes are categorized in several fields, such as UAV-aided power transfer, UAV-assisted Mobile Edge Computing (MEC), and UAV-assisted communications. Alzahrani et al. (2020) analyzed the advantages of the UAV assistance paradigm, such as deploying an aerial energy source to ground network users, deploying flying monitoring support, and establishing a communication bridge between isolated networks. Regarding the integration of security aspects into UAV-assisted networks, Fotouhi et al. (2019) have covered a variety of topics of UAV-assisted networks and their cyber-physical security issues. Furthermore, Nguyen et al. (2021b) comprehensively surveyed UAV-based DC schemes in WSNs while considering different security issues of such solutions. Finally, Altawy and Youssef (2016) discussed a set of issues related to safety, privacy, and security that arise with using civilian UAV-assisted systems and proposed some security requirements for such systems. Baltaci et al. (2021) provided an extensive study on aerial vehicle-based connectivity applications using future communication technologies. Dao et al. (2021) provided a comprehensive study regarding Aerial Radio Access Networks (ARANs) in terms of network design, operational management, performance evaluation models, and data delivery planes.

Besides, with the emergence of diversified UAV-based applications, some surveys on DC approaches have come along. In particular, Mohamed et al. (2020) investigated the potential of involving UAVs in smart city applications and the issues facing such involvement. Hayat et al. (2016) highlighted the requirements of UAV networks for civilian applications and then studied the different UAV communication technologies. Motlagh et al. (2016) analyzed the architecture of UAV-assisted IoT networks and the related challenges. Moreover, Wang et al. (2017) presented the architectures of Flying Ad hoc Networks (FANETs) and provided numerous distributed gateway selection methods and cloud-based stability control strategies. The work proposed in Shakeri et al. (2019) highlighted the main challenges of UAV-assisted cyber-physical systems (CPSs) applications, such as UAV trajectories, UAV energy consumption, and UAV-enabled image processing. Zhang et al. (2020a) studied and discussed the main applications of UAVs in WSNs, based on three categories: (i) auxiliary WSN DC, (ii) collaborative

sensing, and (iii) autonomous deployment of UAVs. Sharma et al. (2020) presented an overview of the recent communication technologies for UAV communication modules. Mishra and Natalizio (2020) discussed the integration of UAVs as new aerial user equipment into 5G cellular networks by touching many emerging technologies, such as non-orthogonal multiple access (NOMA), Reconfigurable Intelligent Surfaces (RISs), and machine learning UAV trajectory optimization.

Another category of surveys mainly analyzes the optimization of UAV trajectory and data transfer during the DC process. For instance, Zeng et al. (2019a) covered several topics related to communication requirements, energy consumption, and channel features in UAV-assisted ground cellular networks. To take advantage of the network architectures of FANETs and nanosatellite swarms, Bacco et al. (2017) conducted a thorough analysis of their features and issues yet to be resolved. Yang et al. (2020) provided a summary of the key issues of UAV-based DC in IoT networks, such as sensor deployment, UAV autonomous navigation, and UAV trajectory planning. Singh and Kumar (2020) discussed the role of trajectory optimization of mobile elements, including UAVs, in the performance improvement of WSNs. This study is based on three main aspects used to formulate the trajectory: (i) trajectory techniques, (ii) applications, and (iii) domains. Xie and Bai (2021) studied the UAV wireless communication area and discussed some UAV-based DC categories, such as UAVs data storage and forwarding, UAV communication relay, and UAV aerial base station (BS). Finally, Wei et al. (2022b) provided a comprehensive review on UAV-based DC over IoT where they studied the different optimization techniques, including AI-based methods applied to optimize the UAV-based DC process.

A comparative study of the above survey articles is presented in Table 1. This study is based on several parameters that highlight the novelties of our survey compared to others. These parameters make up the categories of UAV-based DC approaches that will be investigated in this survey, each of which will be defined in future sections.

1.3. Contributions

Based on the comparative study carried out in Table 7, we distinguished that several surveys have superficially discussed the UAV-based DC concept. Therefore, we can conclude that the topic of UAV-based DC has not been significantly studied over the last few years. The methodology followed in carrying out this survey passed through several steps. First, we collected the most relevant and recent references around UAV-based DC. Second, we proposed a novel taxonomy to classify the collected references based on the used technology and the scenario where the DC process is deployed. Third, we generated a set of learned lessons after describing each category of references to highlight what has been done until now. Finally, we predicted a set of future directions to highlight what to do next and allow scientists to have a clear vision of the perspectives of this research field. This study motivates us to extensively review UAV-based DC schemes and guide scientists in proposing many contributions on this topic. As far as we can tell, this study is the first in-depth and extensive survey of the UAV-based DC concept, which is organized around the following contributions:

- We discuss the UAV-based DC concept, along with various definitions related to the elements of the system. Moreover, we provide a discussion related to the communication links and channel modeling, along with a brief comparative study. Then, we define the most commonly calculated metrics and outline challenges distinguished during the DC process.
- We provide the motivations behind adopting UAVs in DC. A novel taxonomy of UAV-based DC schemes is proposed and classified the collected references based on ten (10) categories: (i) Movement control, (ii) Data transfer, (iii) Time-sensitive, (iv) Applications, (v) Energy-efficiency, (vi) RIS, (vii) MIMO-UAV, (viii) NOMA-UAV, (ix) Security, and (x) WPT.

- We present a detailed overview of existing UAV-based DC schemes. Moreover, we describe the functionality of each proposed scheme along with its purposes, advantages, drawbacks, and addressed challenges.
- From this study, a set of key lessons learned are highlighted, which allows researchers to have a full view of the topic and propose possible solutions to the distinguished gaps.
- We list the main simulation tools that have been used to assess the effectiveness of the studied schemes, along with the primary calculated metrics and the contributions considered in the comparisons.
- We outline future research directions and open issues related to UAV-based DC. Moreover, possible solutions are provided along with recommended references.

1.4. Organization of the survey

To have a clear idea about the remainder of this survey's structure, there are six sections, each of which discusses specific details related to the topic of UAV-based DC, as depicted in Fig. 1.

- Section 2 outlines the background of the UAV-based DC concept along with its set of definitions.
- Section 3 discusses the motivations behind adopting UAVs in the DC process. Then, it provides a novel taxonomy of UAV-based DC schemes.
- The various UAV-based DC systems are discussed in Section 4 along with their goals, benefits, limitations, and learned lessons.
- Section 5 discusses the common myths and critical questions about UAV-based DC.
- Section 6 presents the simulation and evaluation tools of the considered schemes.
- Section 7 outlines issues and potential research directions.
- Section 8 concludes the survey.

Table 2 lists essential acronyms, abbreviations, and mathematical symbols used throughout the survey.

2. UAV-based data collection: Background and challenges

The deployment of UAVs to serve terrestrial devices has proven its worth in fulfilling different tasks, such as wirelessly connecting devices (Zhan and Zeng, 2022), environmental monitoring (Khosravi and Samadi, 2021), and relaying data to the CC (Zhong et al., 2020a). Therefore, the DC concept is a keystone to establishing the interaction between UAVs and terrestrial networks. As illustrated in Fig. 2, there are three main scenarios where UAVs are deployed to support DC: (i) UAV-assisted static nodes, (ii) UAV-assisted mobile nodes, and (iii) UAVs as sensing devices. However, UAV-based DC can be pretty challenging in the sense that some criteria should be satisfied, such as reducing the energy consumption of UAVs to prevent network failure, considering the energy constraint of served devices, and ensuring reliable UAV communications by incorporating efficient communication using existing channel models. In this section, we briefly review the system elements, background, and channel modeling related to the UAV-based DC concept to learn more about how UAVs can assist the DC process. Then, we discuss the deployment challenges of UAVs as data collectors.

2.1. System elements

As depicted in Fig. 2, a typical UAV-based DC concept comprises several elements with different characteristics, deployments, and processes (Xiao et al., 2021b). These elements have a continuous interaction among themselves and complement each other. Each element is clearly described in the following subsections to give the readers comprehensive knowledge and a firm hold on the principles of the UAV-based DC concept.

Table 1

Comparison of existing surveys on UAV-based DC. (✓) shows that the subject has been discussed. (×) shows that the subject has not been discussed. (∂) shows that only a portion of the topic has been covered.

Year	Surveys	UAV control	Data transfer	Time-sensitive Data	DC applications	Energy-efficiency	RIS	MIMO	NOMA	Secure	WPT	Future directions on DC	Description
2022	Wei et al. (2022b)	✓	×	×	×	×	×	×	∂	×	×	∂	UAV-assisted DC for IoT.
2021	Pakrooh and Bohlooli (2021)	∂	∂	×	∂	∂	×	×	∂	×	∂	∂	A service-oriented classification on UAV-assisted IoT.
	Nguyen et al. (2021b)	∂	∂	×	×	×	∂	×	×	×	×	∂	UAV-assisted DC in WSNs.
	Xie and Bai (2021)	∂	∂	×	×	∂	×	×	×	×	×	×	UAV-based DC mechanism over WSNs.
	Baltaci et al. (2021)	∂	∂	∂	×	×	×	×	×	∂	×	×	Connectivity applications of aerial vehicles.
	Dao et al. (2021)	✓	∂	×	✓	∂	✓	✓	✓	∂	∂	×	Aerial Radio Access Networks (ARANs) for 6G.
2020	Zhang et al. (2020a)	∂	∂	∂	∂	∂	×	×	×	×	×	∂	UAV-assisted applications over WSNs.
	Yang et al. (2020)	∂	×	×	×	×	×	×	×	×	×	×	Trajectory optimization schemes in UAV-assisted IoT.
	Mohamed et al. (2020)	×	×	∂	∂	∂	×	×	×	×	×	×	UAVs applications in future smart cities.
	Singh and Kumar (2020)	∂	×	×	×	×	×	×	×	×	×	∂	Mobile elements-based DC in WSNs.
	Alzahrani et al. (2020)	∂	∂	∂	∂	∂	×	∂	×	×	∂	∂	UAV assistance paradigm concept.
	Sharma et al. (2020)	✓	×	∂	∂	∂	×	∂	∂	∂	∂	∂	Future communication technologies for UAV networks.
	Mishra and Natalizio (2020)	✓	×	∂	×	∂	×	✓	✓	×	∂	∂	UAV integration into B5G networks.
2019	Zeng et al. (2019a)	∂	×	×	×	∂	×	✓	×	×	×	×	UAVs Communication for 5G/B5G.
	Fotouhi et al. (2019)	×	×	×	×	∂	∂	✓	×	×	×	×	UAV Cellular Communications.
	Shakeri et al. (2019)	∂	×	∂	×	×	×	×	×	×	×	∂	Cyber-physical applications within multi-UAV systems.
2018	Sekander et al. (2018)	×	×	×	×	∂	×	×	×	×	×	×	UAV-assisted 5G/B5G cellular networks.
	Sonawane and Sujithra (2018)	×	×	×	×	∂	×	×	×	×	×	×	Energy-efficiency of UAV communication network.
2017	Wang et al. (2017)	×	×	∂	×	×	×	×	×	×	×	×	Distributed Cooperative gateway election in UAV networks.
	Bacco et al. (2017)	×	∂	×	×	×	×	×	×	×	×	×	Network architecture for UAV swarms and Nanosatellites.
	Jawhar et al. (2017)	×	∂	×	×	∂	×	×	×	×	×	×	Architecture of UAV-based communication systems.
2016	Altawy and Youssef (2016)	×	×	×	×	×	∂	×	×	×	×	×	Security features of civilian UAVs.
	Hayat et al. (2016)	×	×	∂	✓	×	×	×	×	×	×	×	UAVs for Civilian Applications.
	Motlagh et al. (2016)	×	∂	×	∂	∂	×	×	×	×	×	×	UAV-aided in IoT Services.
Our survey		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	Deep analysis on UAV-based DC.

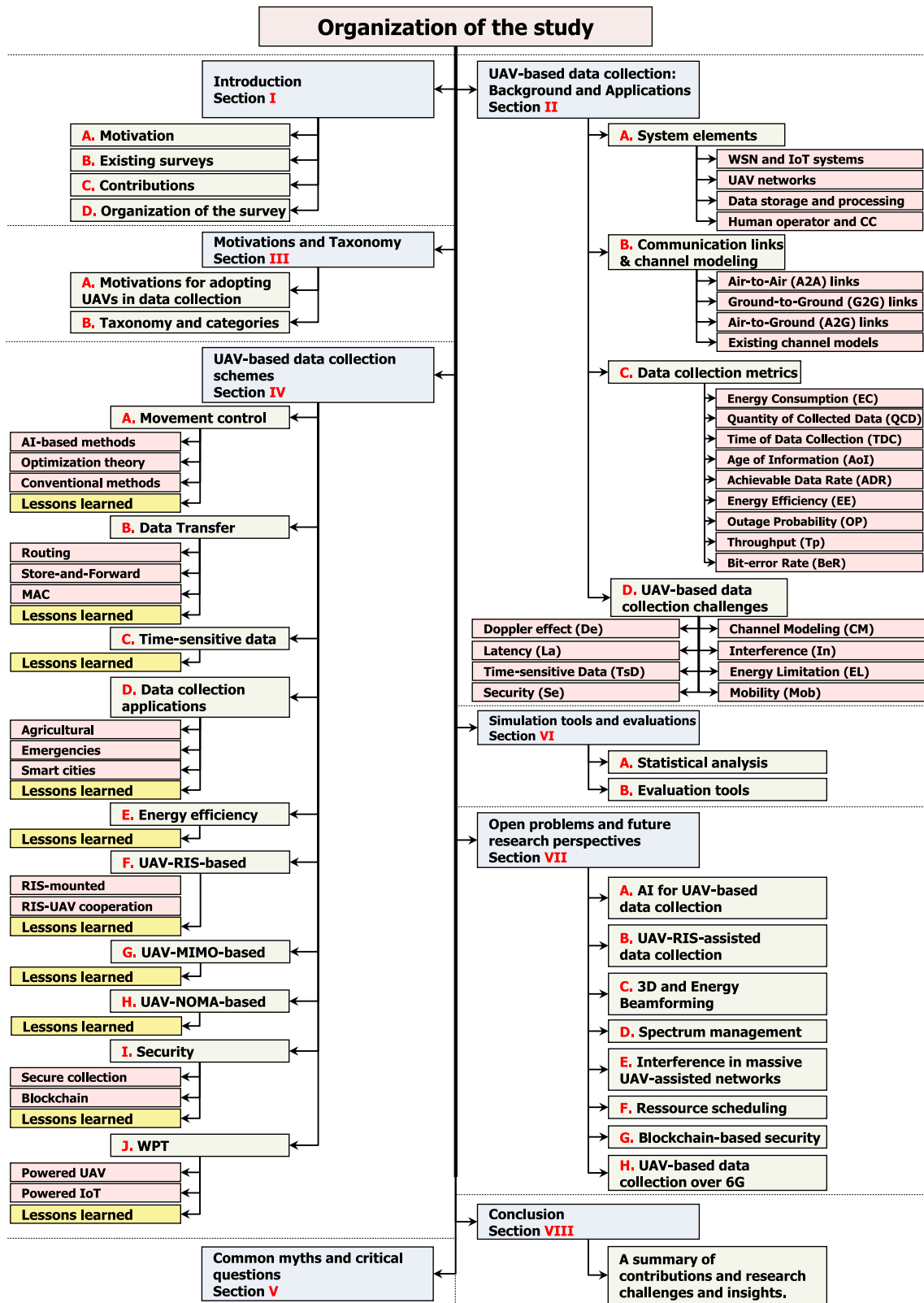


Fig. 1. Organization of the survey.

2.1.1. WSN and IoT systems

As illustrated in Fig. 3, WSN and IoT systems have different architectures, partly due to different operating principles and components. A WSN consists of electronic devices called sensors, spatially dispersed and interconnected through wireless links (Oroza et al., 2021).

Each sensor mainly consists of a battery, storage unit, processor, and transceiver module. A sensor measures the fluctuation of current conditions in its surroundings and wirelessly transmits the sensed data through its transceiver module to a selected sink node, referred to as a gateway, either directly or through other sensors. However, in

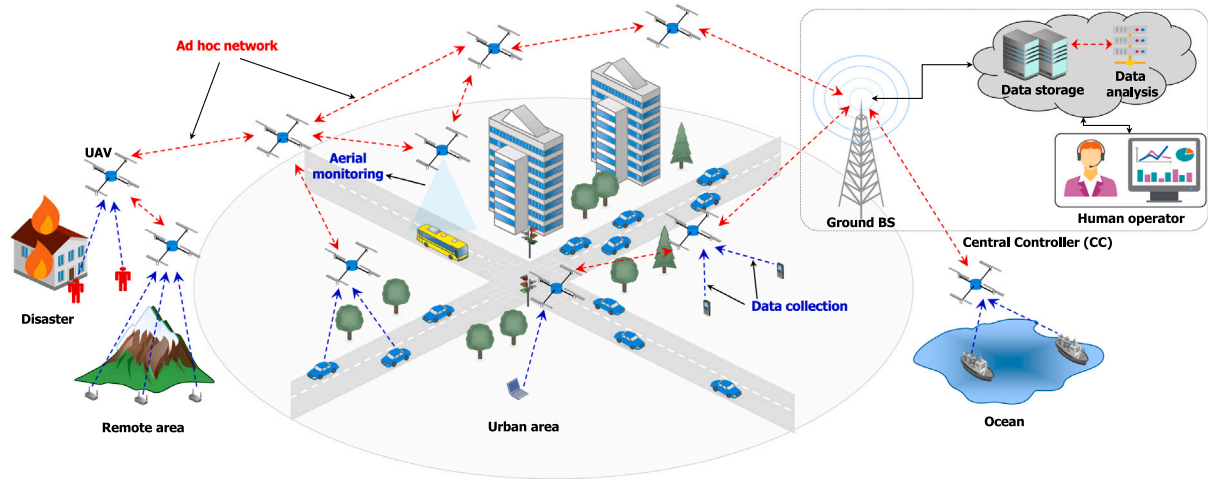


Fig. 2. Application scenarios of UAV-based DC.

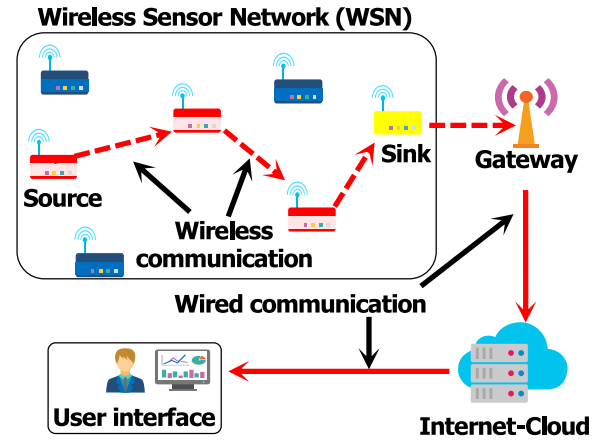
the IoT paradigm, there is a set of smart devices/sensors, also called IoT devices, which can be static or mobile and could be deployed at various locations (Da Xu et al., 2021). Moreover, all such devices could exchange information and be connected based on some protocols to get positioning, control, or smart reorganizations. Also, IoT devices are generally connected directly to on-premises servers or cloud platforms on the Internet, providing them an important number of resources to store and process data centrally (Nguyen et al., 2021a).

The WSN and IoT systems may have many dissimilarities and different functional capacities. Table 3 provides an overview of the technical differences.

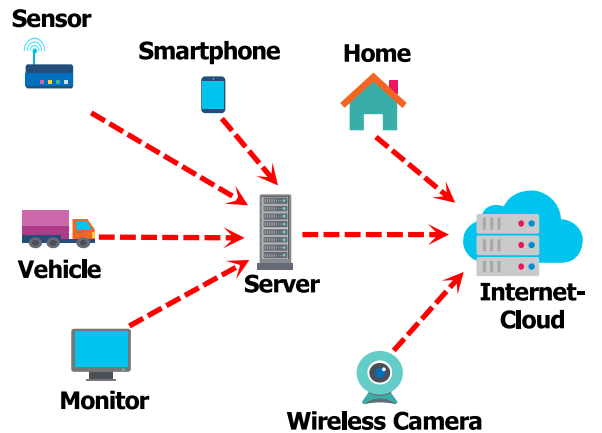
2.1.2. UAV networks

UAV networks are composed of multiple restricted-energy UAVs acting as aerial BSs, relays, edge servers, or even data collectors (Hu et al., 2019). To play these roles, there are many types of UAVs with different equipment, energy capacities, ranges, and shapes (da Silva and da Fonseca, 2023). All these types of UAVs are available in the market, and they support various applications thanks to their sensors, global positioning systems (GPSs), cameras, stabilizers, and wireless interfaces (Radoglou-Grammatikis et al., 2020). Generally, UAVs can be classified into four main categories: (i) fixed-wing, (ii) fixed-wing hybrid, (iii) single rotor, and (iv) multirotor. Indeed, fixed-wing UAVs are made of wings to create lifting force, a propeller to rotate, and a motor to generate power propulsion. This type of UAVs is generally deployed for aerial mapping and line inspection (Licea et al., 2023). As for the fixed-wing hybrid UAVs, they are a little bit different from fixed-wing UAVs in the sense that they combine forward flight with hovering abilities (Sonugur, 2022). This UAV type can be deployed in long-term missions, such as interplanetary travels or monitoring missions. On the other hand, single-rotor UAVs are commonly known as helicopters, which are able to vertically land and take off, and they can carry high payloads and have longer endurance compared to multirotor UAVs (Hong et al., 2021). The cheapest and easiest UAVs to deploy are multirotor UAVs, commonly used for short-term missions. These UAVs are prevalent in the industry and academia because of their low cost, small size, high agility, and fast maneuverability (Zheng and Zhang, 2019). They can be classified into four main subcategories: (i) tricopter, (ii) quadcopter, (iii) hexacopter, and (iv) octocopter. These UAV types are briefly summarized in Table 4.

Usually, UAVs could communicate in an ad hoc manner or through a terrestrial BS according to three main architectures (Baltaci et al., 2021). First, the centralized architecture is used to connect a set of UAVs through a single BS playing the role of relay for inter-UAV communications. Second, the clustering architecture establishes several



(a) WSN system.



(b) IoT system.

Fig. 3. General WSN and IoT architectures.

clusters that only restrict UAV cluster-heads (CHs) from communicating with the BS and allows other UAVs belonging to the same cluster to communicate in an ad hoc fashion. However, the extra-cluster communication should transit through the BS and CHs. Third, the cellular architecture is mainly used in civilian and military applications, which deploys multiple sets of UAVs operating in different cells. In this

Table 2
List of Acronyms/Abbreviations/Symbols.

Label	Explanation
5G	Fifth-generation
6G	Sixth-generation
A2A	Air-to-Air
A2G	Air-to-Ground
AoI	Age of Information
BS	Base station
CC	Central Controller
CH	Cluster-Head
Cm	Channel modeling
CPS	Cyber-Physical System
DC	Data Collection
De	Doppler effect
La	Latency
El	Energy limitation
FANET	Flying Ad hoc Network
G2G	Ground-to-Ground
In	Interference
IoT	Internet of Things
MEC	Mobile Edge Computing
Mob	Mobility
QoS	Quality-of-Service
RIS	Reconfigurable Intelligent Surface
RL	Reinforcement Learning
Se	Security
SWIPT	Simult. Wireless Information and Power Transfer
TDMA	Time Division Multiple Access
Tsd	Time-sensitive Data
UAV	Unmanned Aerial Vehicle
WET	Wireless Energy Transfer
WIT	Wireless Information Transfer
WPCN	Wireless Powered Communication Network
WPT	Wireless Power Transfer
WSN	Wireless Sensor Network
Symbol	Explanation
U_k^t	Update generated by device k at time t
$A_k[t]$	AoI of device k at time t
N_i	Number of times that tool i is used
R	Number of discussed UAV-based DC schemes
EE	Energy Efficiency
EC	Energy Consumption
OP	Outage Probability
T_p	Throughput
BeR	Bit-error Rate
DS	Data successfully received
TD	Total amount of Data
QCD	Quantity of Collected Data
t	Unit of time
T	Duration of DC mission

Table 3
Technical differences of WSN and IoT systems.

Features	WSN system	IoT system
Internet	Not required	Required
Devices	Single kind	Many kinds
Mobility	Occasionally	Very common
Storage	Rarely	Common
Connection	Device-to-Device	Device-to-Internet
Applications	Single application	Many applications

architecture, each cell operates in a certain frequency range to stay away from interference with neighboring cells. The three architectures of UAV networks are illustrated in Fig. 4.

Adopting UAV networks to assist terrestrial networks could provide several benefits (Mozaffari et al., 2019). First, do UAV networks extend the coverage of terrestrial networks? Second, what is the maintenance and deployment cost of UAVs? Third, what are the consequences of UAV failure? Firstly, UAV networks may expand their coverage by dispatching more UAVs and using efficient routing methods. Secondly, UAVs are less expensive to deploy and maintain than large UAVs with heavy payloads and complex hardware. Finally, in the case when a UAV

Table 4
UAV classification and features.

UAV categories		Main specifications	
Fixed-wing		Long-term missions	high speed
Fixed-wing hybrid			hovering
Single-rotor		Mid-term missions, hovering	
Multirotor	Tricopter	Short-term missions	3 propellers
	Quadcopter		4 propellers
	Hexacopter		6 propellers
	Octocopter		8 propellers

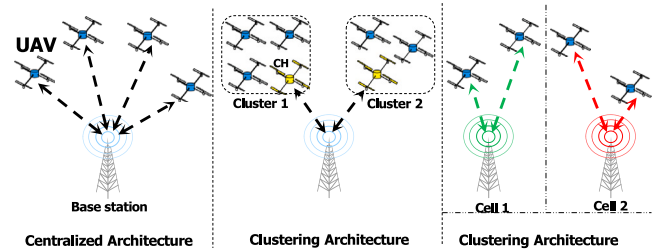


Fig. 4. UAV network networks.

becomes inoperative, the mission of UAV networks can continue with other UAVs.

2.1.3. Data storage and processing system

Such a system consists of a series of resources, such as storage servers, databases, and high-speed network bandwidth (Cui et al., 2021). Moreover, this system is located near the IoT environment and is responsible for receiving large amounts of data and ensuring out-sourced data storage support to devices. This system is also composed of a computation module to perform statistical analysis on the stored data and make decisions accordingly (Awaysheh et al., 2021). The computation module could support different applications executed by devices belonging to the network and having computational tasks, thus centralizing task processing.

2.1.4. Human operator and CC

Most of the time, the human operator is responsible for analyzing, interpreting, and validating the data provided by the processing system (Sacco et al., 2020). Moreover, the human operator is also in charge of preparing the datasets for each part where the training should be performed. As for the CC, it is considered as the whole system, comprising the human operator, data storage and processing system, and BS, which is able to identify the integrity of collected data and support the communication and scheduling between its different modules (Huda and Moh, 2022). Moreover, the CC could also control the trajectories of UAVs for optimizing the DC process (Wei et al., 2022b).

2.2. Communication links and channel modeling

The assistance of UAVs to terrestrial networks for DC purposes requires the support of various types of communication links (Li et al., 2022c). Generally, there are three types of links: (i) Air-to-Air (A2A) links, (ii) Air-to-Ground (A2G) links, and (iii) Ground-to-Ground (G2G) links. Because each connection has unique channel properties, each link has to be modeled uniquely. In Table 5, we provide a brief comparative study of the different links based on various parameters. The following subsections present a comprehensive description of the three types of links, along with some existing models that can be eligible for each kind of link.

Table 5
Measurements of common communication links for UAV-based DC.

Features/Links	A2A	A2G	G2G
Frequency	2.4/5 GHz		2.4 GHz
Throughput	[13, 54] Mbps	[11, 600] Mbps	[0.25, 55] Mbps
Latency	[10, 100] ms	[1, 100] ms	[15, 100] ms
Coverage	Up to 120 m	[120, 250] m	[100, 2500] m
References	Yanmaz et al. (2014), Jimenez-Pacheco et al. (2012), Frew and Brown (2008)	Fadlullah et al. (2016), Huiru et al. (2016), Ho and Shimamoto (2011)	Morgenthaler et al. (2012), Chen et al. (2017a), Trasviña-Moreno et al. (2017), Rhee et al. (2004)

2.2.1. Air-to-Air (A2A) links

The inter-UAV communication is important for networking assistance and UAV-assisted DC. Since UAVs are deployed in the air, the exchange of information between them is carried out through A2A channels. Therefore, the A2A channel is more susceptible to undergoing Line-of-Sight (LoS) propagation in an open environment (Li et al., 2020b). However, the A2A channel can have Non-LoS (NLoS) environment according to the altitude of UAVs and their surroundings in terms of obstacles, such as buildings or other scattered UAVs (Azari et al., 2020). For instance, when UAVs fly at low altitudes in urban environments, the A2A channel may experience a multipath problem due to blockage from nearby buildings. This means that the channel fading of A2A channels is weak, and thus the leading cause of information exchange failure is external interference. Moreover, the high mobility of UAVs makes the A2A channel have a significant Doppler shift.

2.2.2. Ground-to-Ground (G2G) links

In some cases, terrestrial IoT devices and sensors need to communicate between each other or between them and ground BSs. This communication is performed via G2G channels that are characterized by the absence of LoS links between communicating nodes (Kim et al., 2021). Based on the setting in which the terrestrial network is installed, G2G channels can be modeled as a Gaussian channel if it is an open environment (e.g., rural areas) or as a Rayleigh fading channel, otherwise (e.g., urban areas) (Wang et al., 2020c). Moreover, G2G channels could support distances up to 22.5 meters and adopt different IEEE standards at the data link and physical layers according to the applications' requirements.

2.2.3. Air-to-Ground (A2G) links

A2G channels could support connections between IoT devices on the ground and UAVs or between Ground BSs and UAVs. A2G channels are LoS dominated, especially when UAVs operate at high altitudes (Yang et al., 2022c). However, A2G channels are more complex to predict compared to traditional channels, particularly in urban areas where UAV communication is exposed to the negative impact of terrain and surrounding obstacles. According to a recent study (Jiang et al., 2020b), obstructed A2G channels could support communications up to 100 meters with a transmission rate of up to 54 Mbps based on the selected standard.

2.2.4. Existing channel models

According to different surveys (Khawaja et al., 2019; Yan et al., 2019), there are three main types of channel models: (i) deterministic, (ii) stochastic, and (iii) geometry-based stochastic. The deterministic models calculate channel parameters using geometric data while considering the theory of electromagnetic wave propagation (Zhu et al., 2022a). Nevertheless, they demonstrated low efficiency in terms of accuracy and became less popular for A2G scenarios (Tashtan and Ilhan, 2022). Stochastic models are the appropriate choice to reduce the complexity of channel modeling and ensure acceptable accuracy. Indeed, they are established based on the tapped delay line system in which various taps exist. Each tap can supply fading characteristics of multipath. As for the geometric-based stochastic models, they are considered an adequate solution for a 3D geometric simulated environment since they obtain spatial and temporal features of communication channels.

In A2A links, we distinguished that a reduced number of deterministic models are adopted. We mention, as an example, the empirical model (Zhou et al., 2018b) and the pathloss model (Zhu et al., 2021). There are also some geometry-based stochastic models, such as those adopted in Ma et al. (2019) and Zhang et al. (2020c). As for A2G links, some deterministic models are considered, such as the pathloss model (Zhou et al., 2022a), ray-tracing model (Liu and Lau, 2019), and finite-difference time-domain model (Shi et al., 2014). Moreover, some stochastic models are also adopted, such as the Tapped Delay Line (TDL) model (Ding et al., 2017) and the curved-earth two-ray (CE2R) model (Yu et al., 2022). The geometry-based stochastic models are also considered in A2G links as in Cheng and Li (2018), Huq et al. (2022), which are dedicated to UAV-MIMO-based mmWave and Terahertz communications. Furthermore, some other classical models in A2G links are also proposed, such as Uniform Ring Model (URM) (Lee, 1973), Uniform Circular Model (UCM) (Abdi and Kaveh, 2002), and Unified Disk Scattering Model (UDSM) (Borhani and Patzold, 2013). Finally, G2G links also adopt the pathloss and ray-tracing models as deterministic models, e.g., in Ozpolat et al. (2021) and Nuckelt et al. (2013), respectively. Furthermore, geometry-based stochastic models are adopted in G2G links, such as in Wei et al. (2022a), Huang et al. (2022) and Gustafson et al. (2020).

2.3. Data collection metrics

Various metrics are calculated to understand and measure the DC efficiency level. Indeed, in the proposed UAV-based DC contributions, a set of key metrics is often calculated to allow the contributors in this topic to optimize the DC performances and show the improvements induced by the newly proposed contributions. The key metrics are explained as follows.

2.3.1. Energy consumption (EC)

IoT devices and UAVs operate through energy-constrained batteries. Therefore, depleted batteries could result in a failure of the overall system (Zeng and Zhang, 2017; Arabi et al., 2018). For this reason, energy consumption becomes an important metric to estimate the residual energy of batteries, thus allowing IoT devices and UAVs to avoid draining their batteries. Generally, this metric considers several metrics related to IoT devices and UAVs, such as their transmission power, movement, and other external physical parameters. There are many models of energy consumption for UAVs and IoT devices from which researchers could study and adopt for their contributions, such as Zeng et al. (2019b), Khan et al. (2021a) and Andres-Maldonado et al. (2019).

2.3.2. Quantity of collected data (QCD)

This amount is determined by dividing the amount of data DS , generally in bits, successfully acquired at UAVs by the total amount of data TD broadcasted by IoT devices. Certain contributions, such as Kurunathan et al. (2021) and Han et al. (2021), estimate the Packet Losses Rate (PLR) during the DC. PLR metric is considered equivalent to QCD, which aims to have global knowledge about the collected data. The QCD metric is calculated as follows:

$$QCD = \frac{DS}{TD}. \quad (1)$$

2.3.3. Time of DC (TDC)

This time is defined as the duration the UAVs need to complete the DC process over IoT devices (Wang et al., 2021b). This duration is often represented as the number of time-slots required for the UAVs to move and collect data from IoT devices.

2.3.4. Age of information (AoI)

AoI is denoted as the measure of collected information freshness at the receivers (i.e., UAVs), which is often described as the elapsed time between the instant of the current update produced at the data source and its reception by the UAV data collector (Yang et al., 2022b). For example, the AoI of an IoT device k where its last update is generated at U_k^t and successfully sent at time-slot t is expressed as:

$$A_k[t] = t - U_k^t. \quad (2)$$

2.3.5. Achievable data rate (ADR)

This metric is defined as the amount of information in bits successfully delivered per time unit t by IoT devices to UAVs (Cao et al., 2021b). This metric is often optimized to allow UAVs to collect a significant quantity of information per unit of s/Hz, and thus reduce the DC completion time. The ADR metric is often calculated as follows:

$$ADR = \frac{QCD}{t}. \quad (3)$$

2.3.6. Energy efficiency (EE)

Since most proposed UAV-based DC solutions tried to reduce the energy consumption of the whole system and especially of the involved devices (i.e., UAVs and IoT devices), the metric of energy efficiency (EE) is often calculated to show the performance efficiency in terms of energy. This metric is defined as the ratio of aggregated data, generally in bits, to the total energy consumption of UAVs or IoT devices, or both during the whole period of mission time T . Usually, it is calculated in bits/Joule using the following formula:

$$EE = \frac{QCD}{EC}. \quad (4)$$

2.3.7. Outage probability (OP)

This metric is commonly calculated in different UAV-based DC solutions, which allows for avoiding imminent disconnection between UAVs and IoT devices. It is defined as the probability P that ADR falls below a specific threshold τ . This metric can be calculated as follows:

$$OP = P(ADR < \tau). \quad (5)$$

2.3.8. Throughput (Tp)

Throughput (Tp) metric can be defined as the total amount of data in bits successfully offloaded to the UAV data collector during the mission period T . Indeed, Tp has roughly the same definition as ADR, but Tp is totally different from it in the sense that it refers to the overall effective data rate by considering the different constraints of the connection, such as protocol inefficiencies and overhead. The system throughput is defined by:

$$Tp = \frac{QCD}{T}. \quad (6)$$

2.3.9. Bit-error Rate (BeR)

The Bit-error Rate (BeR) is frequently calculated in evaluating the proposed UAV-based DC solutions due to the different physical phenomena that negatively impacts the communication channel between UAVs and IoT devices. Indeed, the BeR is defined as the ratio between the number of bit errors Err to the total number of bits transmitted TD during the studied time interval T . The BeR is calculated as follows:

$$BeR = \frac{Err}{TD}. \quad (7)$$

2.4. UAV-based data collection challenges

Although the importance of UAVs is growing, their deployment as data collectors is still considered a challenging task due to various constraints (Liu et al., 2020f). However, understanding these constraints makes it possible to propose adequate approaches that could be adapted to any deployment environment. In this subsection, we will briefly discuss some key challenges and limitations that UAVs face when they assist ground networks during the DC process:

2.4.1. Channel modeling (Cm)

Owing to the heterogeneous nature of UAV-assisted networks, channel modeling is often constrained by Non-Line-of-Sight (NLoS) conditions existing between UAVs and terrestrial IoT devices, making A2G channels more vulnerable to blocking (Zhu et al., 2022a; Huang et al., 2021c). Moreover, the accurate modeling of the channel requires considering several parameters, such as the elevation angle and height of UAVs, propagation environment, existing obstacles, and interferences. It is worth noting that more extensive simulations and measurements are needed to put on the field a proposed channel model.

2.4.2. Interference (In)

More interferences might be caused when UAVs are deployed to assist terrestrial networks. Indeed, these interferences are often generated when IoT devices transmit their data simultaneously over the same communication bands (Mkiramweni et al., 2018). Moreover, they can be generated when UAVs share the same channel and relay the collected information to the CC (Yang et al., 2021). Generally, interferences lead to different crucial issues, such as possible communication failures, more energy consumption to retransmit the data, and increased duration of the DC mission. Consequently, interferences should always be carefully considered in channel modeling.

2.4.3. Energy limitation (EL)

Since the energy source of UAVs comes mainly from their rechargeable batteries, their mission duration is significantly impacted by their restricted battery capacity (Abeywickrama et al., 2018). In fact, UAV energy consumption depends strongly on their mobility, communication ability, and the energy consumed by their integrated circuits. Moreover, IoT devices are also considered energy-constrained devices, where their energy consumption depends on the periodicity of their data transmission towards UAVs and their possible occasional movements (Zhong et al., 2020b). These facts raise a significant challenge that strongly limits the operating time of the DC operation time. Consequently, it is vital to extend the duration time of the DC process or even to use advanced charging technologies to ensure a continuously operating process.

2.4.4. Mobility (Mob)

To ensure an efficient DC process, UAVs need to continuously move around terrestrial IoT devices and maintain permanent connectivity with neighboring UAVs to relay collected information (Gul et al., 2022). However, finding optimal trajectories of UAVs is considered a challenging problem due to different constraints and real-world practical restrictions, such as the UAV energy consumption, the existing obstacles, the IoT devices' occasional movements, and the movements of other UAVs (Chen et al., 2020). As a result, adopting dynamic trajectory control approaches for UAVs is more appropriate for ensuring a long connection lifetime while keeping adequate coverage of the target area of interest.

Table 6

Summary comparison of the DC process with and without UAVs.

Parameters	Data collection without UAVs	UAV-based data collection
Speed	DC takes more time.	With UAVs, DC can be done quickly if the trajectories of UAVs are well-optimized and adapted to the IoT environment.
Energy consumption	IoT devices consume more energy to transmit their generated data to the collector.	UAVs fly closer to IoT devices, allowing them to transmit their data with a reduced transmission power, thus consuming less energy.
Mobility	The occasional movements of IoT devices constrain data collectors.	The mobility of UAVs can easily be adapted to the mobility of IoT devices.
Interference	Difficult to cope with interferences generated by IoT devices.	Interferences can be mitigated if different links are modeled by considering interferences and mechanisms to avoid them.
Flexibility of deployment	DC is challenging to be done in complex areas.	UAVs can be deployed in any difficult accessing areas to make DC.
Security	Several security schemes are developed, which are easily breakable.	Require more effort to design security approaches while involving the characteristics of UAVs.
IoT charging	Fixed charging stations are not always efficient for permanently charging IoT devices.	UAVs can act as aerial charging stations and supply IoT devices with energy.
Real-time Applications	Data collectors do not care about the expiry deadlines of collected data.	UAVs could be well-exploited for collecting and transmitting data to the CC while considering their expiry deadlines.

2.4.5. Security (Se)

Although the UAV-assisted terrestrial network concept has proven its efficiency in significantly improving the DC process, the transmission of data towards UAVs is prone to interception by potential malicious eavesdroppers owing to wireless channels' broadcast nature (Huang and Savkin, 2021). This security breach poses a risk to data security and confidentiality. Moreover, the DC process is exposed to other kinds of threats, such as man-in-the-middle, spoofing, and unauthorized modifications in the storage of collected data, which might lead to uncertainty about data integrity (Basan et al., 2021). Therefore, different security mechanisms are required to ensure a certain degree of reliable DC.

2.4.6. Time-sensitive data (Tsd)

In specific scenarios, IoT devices are deployed to continuously collect time-sensitive data, such as temperature, obstacle information, or monitoring data. These data are supposed to convey useful information when uploaded within a given target time, beyond which they lose meaning and become irrelevant (Gao et al., 2023b). The distinguished challenge is that UAVs collect data without considering the expiry deadline of each collected data. Therefore, to avoid collecting irrelevant data, UAVs should make the DC process while having global knowledge of the expiry deadlines of each data.

2.4.7. Latency (La)

The latency is defined as the total amount of time taken to relay information from the instant of collection until its arrival at the CC. This metric can be estimated for two cases. The first case is communication latency, where UAVs should establish a cooperative relay scheme instead of transporting collected data based on their movements to the CC to reduce this period of time (Zhou et al., 2020a). The second case is what we call computation latency, which is defined as the time required for computations to process the collected data (El Haber et al., 2021). Therefore, end-to-end connection schemes considering computation latency should be designed to relay collected reliably between the UAV data collectors and their respective CCs.

2.4.8. Doppler effect (De)

The Doppler effect can be described as the frequency shifts noticed in transmitted waves when the source and destination are in relative motion (Chang et al., 2023). The challenge in the UAV-based DC process is that the Doppler effect causes significant inter-carrier interference, resulting in the delayed collection of channel state information, severely degrading the communication capacity between UAVs and IoT devices. Therefore, to avoid such issues, different mechanisms should be considered to ensure a certain quality of the established channels.

3. Motivations and taxonomy

After discussing the background and challenges of the UAV-based DC concept in Section 2, we are now ready to go over the motivations behind adopting UAVs as data collectors. In this section, we begin by outlining the broad reasons for adopting UAVs in the DC process. Then, we discuss the proposed taxonomy of UAV-based DC and describe how the different categories are defined.

3.1. Motivations for adopting UAVs in data collection

Table 6 summarizes the comparison of classical DC without UAVs and UAV-based DC. To be clear, IoT devices or sensors have two possibilities to upload their data to the CC: (i) through a direct link towards the CC or (ii) based on UAV data collectors. The first option is costly in terms of IoT energy consumption, where IoT devices need to invest more transmission power to upload their data to the CC that can be located far from them, and thus quickly draining their batteries. As for the second option, By employing UAVs as data collectors, DC can be made more quickly, reliably, and efficiently. In addition, the mobility of IoT devices and the complex areas where they are deployed can now be overcome by UAVs despite their restricted resources (Li et al., 2022b). The primary motivations for using UAVs in DC are highlighted in the following:

- When increasing the speed of UAVs, some parameters, such as residual energy levels of UAVs and the De, should be taken into account when collecting data (Lin et al., 2022). Since UAVs are characterized by their high agility and flexibility, the DC time will be considerably reduced. Moreover, the more the speed of UAVs increases, the more the number of collected bits increases.
- IoT objects are usually energy-constrained devices and hence consume more energy when transmitting their sensed data (Mondal et al., 2021). By employing UAVs as data collectors, the DC process will be carried out closer to IoT devices reducing the transmitting power of IoT devices, and thus reducing their energy consumption.
- UAVs are known for their flexibility of deployment and dynamic movements (Wu et al., 2022b). As compared to fixed data collectors, UAVs can adjust their movements according to both the occasional movements of IoT devices and the IoT devices that want to upload their data.
- A UAV network usually coexists with other networks, e.g., IoT environment (Wu et al., 2021b). With the assistance of the UAVs' built-in mobility, UAVs have the ability to deploy an interference avoidance mechanism (Hosseinalipour et al., 2019). Moreover, other channel access mechanisms can be adopted to mitigate the interference impact (Mohamed et al., 2019). These mechanisms increase the reliability of UAV-based DC.

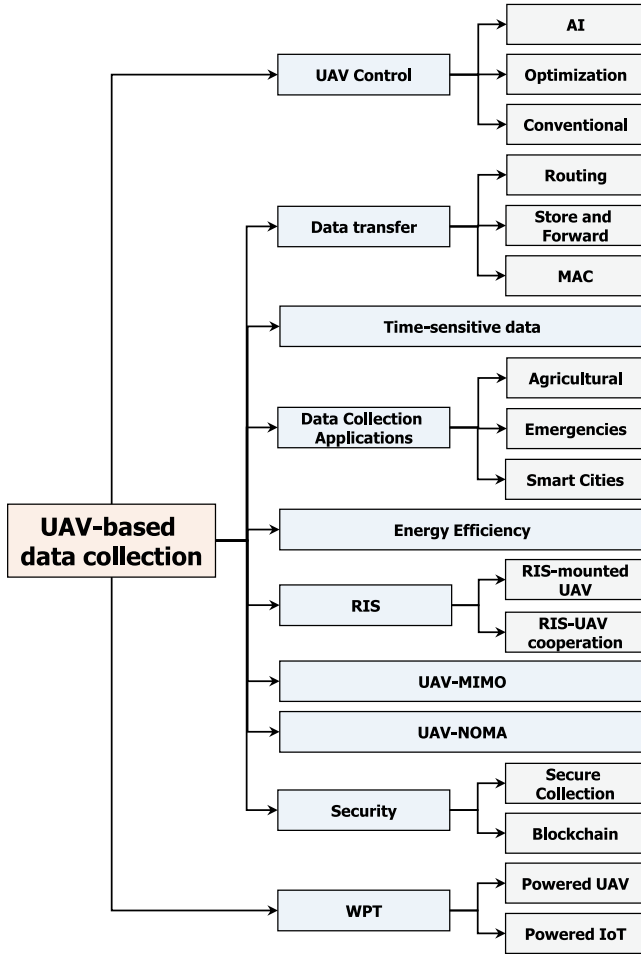


Fig. 5. Taxonomy of UAV-based data collection.

- IoT devices can be deployed over various complex, and harsh environments, which are difficult to be accessed by terrestrial data collectors (Zhao et al., 2019). By integrating UAVs in these environments, DC becomes easier while overcoming the complexity of the deployment area.
- In the scenario where fixed data collectors are dispatched to gather data from terrestrial IoT devices, potential eavesdroppers could intercept the communications between them (Wang et al., 2019c). However, with the help of UAVs, legitimate channels could be established owing to the proximity of UAV data collectors and IoT devices (Lei et al., 2019). In another scenario, UAVs could play the role of jammers to prevent the attacks of eavesdroppers and other kinds of attacks (Wang et al., 2019a).
- IoT devices are usually battery-powered and need to be permanently supplied with energy for the proper functionality of their transmission process towards UAVs (Li et al., 2018). To overcome this challenge, UAVs could act as energy transmitters and support IoT devices with energy before making the DC process (Carrillo et al., 2021).
- Some real-time applications require a specific latency restriction or a given degree of freshness of collected data from IoT devices to ensure their proper functioning (Joseph et al., 2020). UAVs have demonstrated across several contributions (some of them will be described in Section 4.3) that they have the ability and flexibility to satisfy these requirements.

3.2. Taxonomy and categories

The taxonomy of our survey is carried out based on three factors: (i) the used technology, (ii) the adopted method, and (iii) the deployment environment. In Fig. 5, there are ten branches, each corresponding to a category describing the adopted mechanism used by UAVs to make the DC process. Now, we discuss each contribution in each branch in detail in Section 4.

4. UAV-based data collection schemes

This section will pay close attention to the different recently proposed UAV-based DC solutions according to the categories to which they belong (see Fig. 5). First, in each category, we select the most relevant references to discuss based on their popularity in terms of citations from Google Scholar source. Then, an overview of the key lessons learned from each category is presented. Finally, after discussing the different references and their learned lessons, we incorporate a brief comparative study in each category, which comprises objectives, advantages, drawbacks, and addressed challenges (see Section 2.4) for each proposed solution.

4.1. Movement control in UAV-based data collection

Optimizing the trajectories of UAV data collectors could significantly facilitate the DC process (Chu et al., 2022). These trajectories could be optimized based on different parameters, such as the IoT devices' locations and CC, to fulfill the optimum throughput, maximize the amount of collected data, and avoid interferences that may occur (Liu and Zheng, 2022). Three great categories are distinguished in the literature: (i) AI, (ii) Optimization Theory, and (iii) Conventional methods, which are trying to optimize the trajectory of UAV data collectors according to their working principle (c.f., Fig. 6). To completely cover these three categories, the different contributions related to each category are summarized in Table 7.

4.1.1. AI-based methods

In this category, UAVs are deployed to quickly adapt their movements to the environment and its possible dynamics. The UAVs could adopt the reinforcement learning (RL) methods that learn from the environment and move based on the environment's feedback. For instance, in Li et al. (2020c), a single UAV is deployed to carry out a DC process over a group of IoT devices performing environment sensing. The authors used a Deep Q Network (DQN) algorithm for optimizing the UAV trajectory and its DC scheduling with the primary purpose of reducing the packet losses due to both fading of aerial channels and buffer overflows at the IoT devices. The authors of Xiong et al. (2020) used a DQN algorithm to adapt the trajectory of a single UAV data collector to the dynamism of the IoT environment and maximize long-term utility. Indeed, the UAV is wirelessly powered at specific locations and collects data from IoT devices. Oubbati et al. (2021) adopted a multi-agent DQN to control the trajectories of two UAVs acting as energy transmitters and data collectors, respectively. This approach seeks to reduce the total amount of energy used by the system and enhance the timeliness of collected data from IoT devices. To increase the quantity of information gathered from IoT devices under the constraint of the traveled distance, Nguyen et al. (2022a) optimized the 3D UAV trajectory using a dueling DQN. This scheme takes into account the realistic physical constraints of communication. Wang et al. (2021a) optimized a single UAV's trajectory using a twin-delayed deep deterministic policy gradient (TD3) under the restrictions of realistic throughput and flying mobility.

UAV data collectors could also adopt supervised and unsupervised learning methods to make the DC process. Indeed, supervised learning is a machine learning algorithm that maps a pair of input-output based on examples of known input-output pairs (Bithas et al., 2019). For

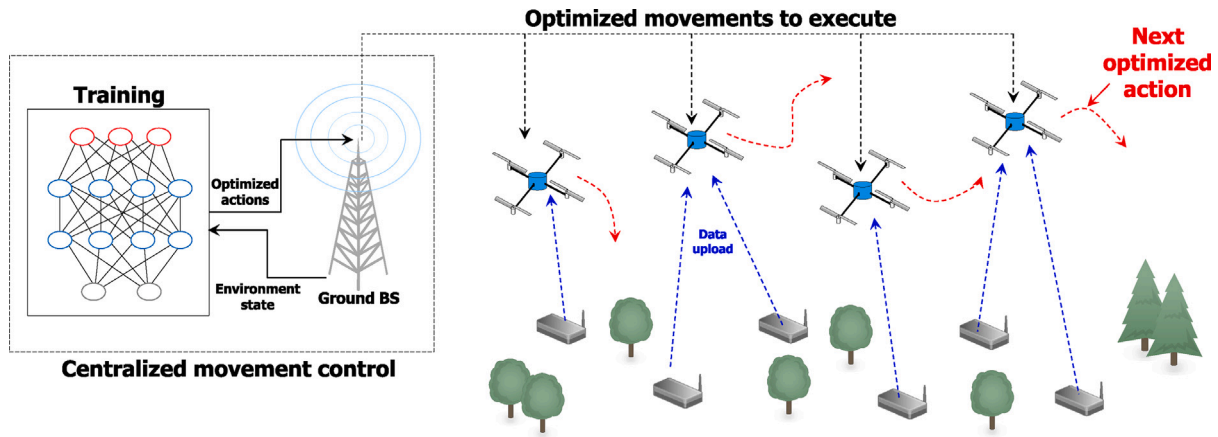


Fig. 6. Centralized movement control optimization in UAV-based DC.

instance, Li et al. (2019b) adopted two algorithms, named group-based trajectory optimization algorithm (GTOA) and segment-based trajectory optimization algorithm (STOA), to solve an optimization problem. This problem consists of optimizing the UAV's trajectory, altitude, and speed while improving the connectivity links with IoT devices to speed up the DC mission. In Chen et al. (2019b), the authors proposed the Direct Future Prediction (DFP) model based on the supervised learning concept. The objective of this model is to optimize multiobjective tasks, which are improving the UAV charging process and maximizing the QCD and their value of information. As for unsupervised learning, it is also a machine learning algorithm, which is different from supervised learning where data is not provided with labels (Hoang et al., 2019). An example of unsupervised learning includes K-Means Clustering, used in the system presented in Yang et al. (2023), where the authors proposed a UAV-based RGB image collection based on unsupervised learning methods. The objective is to use and evaluate the performances of these methods to map various small geographical areas of pit wall sections in two mine sites.

4.1.2. Optimization theory

Several optimization algorithms could be adopted to optimize the UAV trajectory and various other parameters depending on the scenario. In this sub-section, we discuss the main adopted optimization methods according to the following sub-categories:

Dynamic programming (DP)

It is defined as a programming method that transforms a decision-making problem is divided into multiple sub-problems (Abubakar et al., 2023). Then, the sub-problems are solved to find the overall solution for the initial decision-making problem. For example, Gong et al. (2018) deployed a UAV-based DC system and studied the impact of UAV speed on the DC process. Indeed, they used the DP method to optimize the UAV speed and IoT devices' transmission power with the objective of reducing the UAV flight time. It was distinguished that the optimal speed depends mainly on the amount of data to be collected, the transmission power of devices, and the distance that separates the UAV from IoT devices. In Zhou et al. (2018a), a UAV-assisted mobile crowd-sensing system is deployed over a static IoT environment. The purpose of this system is to calculate the optimal UAV energy-efficient trajectory and optimize the sensing task assignment. The two problems are solved using the DP method and genetic algorithm, all applied using an iterative matching algorithm.

Branch and bound (BnB)

It is an iterative approach that aims to split the global problem into smaller subsets of problems (Zhang et al., 2019a). It is typically employed to solve combinatorial optimization problems. For instance, Lu

et al. (2022b) deployed a single UAV to perform the DC process over a set of IoT devices. The idea behind this system is to optimize the UAV trajectory and radio resource allocation to define the minimal UAV energy consumption required to complete the DC process. This multiobjective problem is solved using the BnB method under mild conditions. Benmad et al. (2021) studied a UAV-based DC system over a powered IoT environment. The purpose was to minimize the UAV flight time while serving all IoT devices with the required energy that allows them to transmit their sensed data. The BnB method is used in the simulation part to optimize the UAV trajectory as well as the charging process of IoT devices.

Successive Convex Approximation (SCA)

It is applied to solve a general optimization problem. To do so, a convex upper bound approximation is constructed for a nonconvex function (Sun et al., 2020). This method is applied by Zhan and Zeng (2019a), where they deployed multiple UAV data collectors to collect data over multiple IoT devices. They tried to establish a cost balance between the energy consumption of UAVs and IoT devices by dividing this objective into two sub-problems: (i) devices' wake-up schedule and UAV trajectory optimization and (ii) transmission power optimization of devices. These two sub-problems are solved by applying the SCA technique that converges to a solution satisfying the Karush-Kuhn-Tucker (KKT) conditions. The authors of Shen et al. (2021b) deployed multiple UAV data collectors above an IoT environment. An optimization is carried out to minimize the duration of the DC process and the number of UAV-based DC operations. The UAV hovering location and trajectory are divided into sub-problems to solve. The problem of hover locations of UAVs is transformed into a convex problem and solved by the SCA technique. As for the UAV trajectory problem, a modified ant colony optimization (MACO) algorithm is adopted to solve it.

Matrix Completion (MC)

It is adopted in many applications, including UAV-based DC applications, where the approach tries to recover the missing data and build the entire matrix (Kortas et al., 2020). Based on this strategy, MC can be applied to a UAV-based DC system by collecting a certain QCD, and after that, each row and column of the matrix will be filled with the collected data. Then, MC is called to optimize the corresponding parameters. This method is adopted by the authors of Liu et al. (2019b) to reduce the energy consumption of the deployed UAV data collector and allows it to minimize the redundancy of collected data. Indeed, a matrix is used to recover the DC points in the covered area, and then an optimal UAV path is generated according to the DC points and the considered constraints.

Table 7

Summary of contributions of movement control in UAV-based data collection.

		Objective	Advantage	Drawback	Cm	In	El	Mob	Se	Tsd	La	De
AI methods	Li et al. (2020c)	Minimizing the packet losses during the DC process by optimizing the trajectory and DC scheduling of a single UAV using DQN.	Reducing the packet loss by over 51%.	Discrete actions of the UAV data collector and its energy consumption are not considered.	√	×	√	√	×	×	×	√
	Xiong et al. (2020)	Maximizing the lifetime of the UAV data collector by wirelessly powering it from different geographical locations allows it continuously gather data.	Achieving improved wireless energy and data transmission methods in terms of wider long-term use of the UAV.	Not adapted to real-time applications where time-sensitive data have to be collected as quickly as possible.	×	×	√	√	×	×	×	√
	Oubbati et al. (2021)	Increasing the time-sensitivity of collected information, minimizing device power transmission, and reducing the UAV energy consumption.	Maximizing the gain in terms of freshness of collected data and energy transfer towards IoT devices.	Did not consider both the continuous mobility of UAVs and Interferences generated during the DC process.	√	×	√	√	×	√	×	×
	Nguyen et al. (2022a)	Maximizing the amount of collected data while considering the UAV's limited flight time and the physical constraints of communication.	Schemes can be adapted to any complex environmental settings.	Neglecting the residual energy levels of IoT devices.	√	√	√	√	×	×	×	×
	Wang et al. (2021a)	Efficient trajectory optimization of data collection from multiple IoT ground nodes using twin-delayed deep deterministic policy gradient (TD3).	Minimizing DC completion time under flight movement and throughput constraints.	The system is based on a single UAV while restricting its movement to 2D space, which is challenging to cover a large area.	√	×	√	√	×	×	√	√
	Li et al. (2019b)	Minimizing the DC completion time.	Completion of DC reliably and with a high probability of LoS links.	The UAV energy consumption is not calculated and not considered.	√	×	×	√	×	×	√	×
	Chen et al. (2019b)	Increasing the QCD along with its value of information while ensuring the charging process of the UAV data collector.	Maximizing the value of information collected at each time slot compared to classical RL techniques.	The charging process is not considered in the optimization method.	×	×	√	√	×	×	×	×
Yang et al. (2023)	Investigating UAV-based RGB images collection for pit wall mapping.	Providing better performances in terms of mapping.	Inducing high convergence time and a huge amount of information required for learning.	×	×	×	×	×	×	×	×	
Optimization Theory	Gong et al. (2018)	Minimizing the UAV flight time while ensuring the completion of the DC mission.	Ensuring a sufficient residual energy level for the UAV so that it can move at its maximum speed while maintaining a certain amount of data to be collected.	The visit order of IoT devices to make DC is not optimized.	√	×	×	√	×	×	×	×
	Zhou et al. (2018a)	Optimizing the UAV trajectory and DC task assignment under the constraint of energy efficiency.	Reducing considerably the UAV energy consumption and maximizing the total profit of the system.	The region's division into subregions DC is not discussed. This division can be unfair.	×	×	√	×	×	×	√	×
	Lu et al. (2022b)	Optimizing the UAV trajectory and resource management for minimizing the UAV energy consumption during the whole mission period.	Reducing significantly the UAV energy consumption required for the DC process.	Since IoT devices can transmit their sensed data simultaneously, the problem of interference is not considered.	√	√	√	√	×	×	×	√
	Benmad et al. (2021)	Minimizing the UAV-based DC process while transferring the required energy level to IoT devices.	Optimizing the charging process of IoT devices while minimizing the DC completion time.	The DC completion time is not optimized enough due to the pre-charging process.	√	×	√	√	×	×	×	×
	Zhan and Zeng (2019a)	Reducing the energy consumption of both UAVs and IoT devices while ensuring the smooth running of the data collection process.	Achieving a cost balance in terms of energy consumption of both UAVs and IoT devices.	Some generated data cannot be transmitted due to IoT devices' failures caused by their lack of energy. Also, the DC completion time is not considered.	√	√	√	√	×	×	×	×
	Shen et al. (2021b)	Minimizing the UAV-based DC operations and DC completion time under the constraints of UAV energy consumption and transmission schedule of IoT devices.	Converging quickly to an optimal solution with better performance compared to other benchmark methods.	The collisions between UAVs are not considered, which can distort the whole system in case they occur.	√	√	×	√	×	×	×	√

(continued on next page)

Other methods

Since we considered a limited number of methods in this subsection, we briefly discuss some other methods applied to optimize

the UAV trajectories. For instance, Samir et al. (2019) derived a non-convex optimization problem to have an optimal UAV data collector trajectory and resource allocation, which maximizes the quantity of

Table 7 (continued).

	Objective	Advantage	Drawback	Cm	In	El	Mob	Se	Tsd	La	De
Conventional methods	Liu et al. (2019b)	Optimizing the UAV trajectory decreases energy consumption and reduces redundant data.	Achieving lower UAV energy consumption and data redundancy.	The energy consumption of IoT devices is not considered.	×	×	✓	✓	×	×	×
	Samir et al. (2019)	Maximizing the number of served IoT devices.	Increased more gain in terms of the number of IoT devices supported (average improvement of 10%–50%).	The energy consumption of IoT devices is neglected.	✓	×	×	✓	×	✓	×
	Liu et al. (2020d)	Minimizing the movement trajectory of the UAV while collecting the maximum of data.	Minimizing data redundancy during the DC process and energy consumption of the UAV data collector.	Ignoring the security aspect of the DC process and missing energy consumption balancing between IoT devices.	×	×	×	✓	×	×	×
	Zhan and Zeng (2019b)	Minimizing the DC completion time under the constraint of an appointed amount of uploaded data.	Delivering significant performance gains compared with the benchmark schemes.	The UAV energy consumption is not considered.	✓	×	✓	✓	×	×	×
	Goudarzi et al. (2019)	Planning the most optimized path should gather as much information as possible.	Decreasing the route length when the amount of collected data becomes sufficient.	The energy usage of IoT devices is not considered.	×	×	×	✓	×	×	✓
	Gao et al. (2023a)	Ensuring an obstacle-avoidance flight in indoor environments using EDT.	A cost-effective method that provides high safety performances.	Adequate only for a single UAV deployment.	×	×	×	✓	×	×	×
	Qayyum et al. (2021)	Providing a timely DC over randomly distributed IoT devices to support real-time farming applications.	Outperforming competitors that belong in the same category in terms of energy efficiency, networking delay, and the number of generated overhead.	Generating high delays in clusters' formation.	×	×	✓	✓	×	×	✓
	Tao et al. (2019)	Deploying multiple UAVs for timely DC while reducing their energy consumption.	Optimizing the UAV trajectory while reducing its energy consumption. It can also be generalized to the deployment of multiple UAV data collectors.	It is not appropriate for dynamic IoT environments.	×	×	✓	×	×	×	×
	Wang et al. (2020d)	Optimizing the flight time of multiple UAVs to reduce the DC completion time.	Reducing significantly the DC completion time.	Did not ensure DC fairness between UAVs especially when IoT devices are grouped in a specific location.	✓	✓	×	✓	×	×	×
	Pěnička et al. (2019)	Establishing a short and collision-free path for DC using UAV.	Demonstrating computational benefits in a very dense roadmap.	The experimentation is carried out using a single UAV, and no generalization is made.	×	×	×	×	×	×	×
	Asadi et al. (2020)	Providing efficient navigation in indoor environments in the presence of obstacles.	Demonstrating good performances in terms of visual data collection and autonomous navigation.	The experiment is based on a straightforward scenario, which cannot evaluate the proposed contribution properly.	×	✓	✓	✓	×	×	×
	Tazibt et al. (2018)	Optimizing the UAV trajectory and reducing the DC completion time while considering the existing obstacles.	Reducing the DC completion time and distance traveled by the UAV data collector.	The energy consumption of the UAV is not considered.	×	×	✓	×	×	×	×

IoT devices supplied, while considering the deadline constraints of data. This approach takes into account the UAV energy consumption. The work in Liu et al. (2020d) proposed a matrix-completion-based Sampling Points Selection (SPS) to reduce the amount of redundancy during the DC process and optimize both the UAV trajectory and its energy consumption. The authors of Zhan and Zeng (2019b) optimized the trajectories of multiple UAVs using min-max multiple Traveling Salesman Problem (TSP). This approach aims to reduce the maximum completion time under the constraint of a fixed quantity of supplied data while taking energy usage of IoT devices using the wake-up scheduling mechanism. In Goudarzi et al. (2019), the authors optimized the UAV data collector's trajectory based on the TSP algorithm. This scheme aims to increase the UAV's flight time while collecting no redundant data and increasing the packets' delivery.

4.1.3. Conventional methods

Graph-theory-based methods are used to optimize the movement control of UAV data collectors. In this section, we discuss the main adopted methods in this category, where the deployment environment of UAVs is modeled as a graph, and the optimal paths of UAVs are then extracted.

Euclidean distance transform (EDT)

This method is considered as a greedy-path-planning technique (Chen et al., 2022). Its objective is to calculate at each time the distance that separates the UAV from the closest obstacle. This process is called the exploration completion logic. When an obstacle is met, the UAV tries to calculate the closest distance to the next free zone. This process is called a dead-end situation. For instance, in Gao et al. (2023a), a UAV-based indoor facility DC and scene reconstruction system is proposed. The objective of this system is to achieve fully autonomous inspection while considering the complexity of indoor environments in terms obstacles and localization. Here, the EDT algorithm is used to obtain the closest distance between the UAV and obstacles and thus ensure a certain level of flight safety.

Dijkstra algorithm

It is defined as a path-planning method used mainly to calculate the shortest path between a given source point to multiple destinations (Prasad and Ramkumar, 2022). The optimal path calculation is based on the cost of different computed paths. For instance, Qayyum et al. (2021) proposed a DC mechanism using UAVs for smart farming real-time applications. This mechanism tries to organize IoT devices in

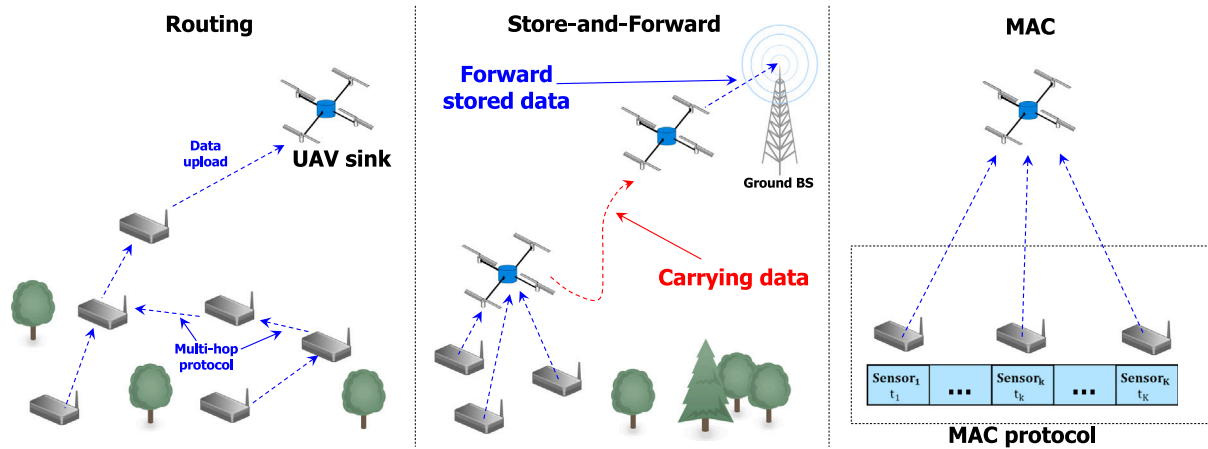


Fig. 7. Data transfer methods in UAV-based DC.

the form of clusters. Then, using the parameters of energy efficiency and networking delay, UAVs adopt the Dijkstra algorithm to find the optimal trajectory for DC.

A-star algorithm

It is also widely adopted in UAV-based DC, which can be considered as an extended version of the Dijkstra algorithm where an expected cost index is added as a parameter to calculate the shortest path between two points (Zhang et al., 2020b). For instance, Tao et al. (2019) proposed a multi-UAV-based DC system deployed in a large-scale IoT environment. The authors adopted the A-star algorithm to define the flight path for each UAV across different IoT devices. Furthermore, to optimize the functioning of the A-star algorithm, they also consider an efficacy function considering many features, such as energy consumption, DC time, and data priority.

Voronoi diagram method (VDM)

It is another widely adopted path-planning algorithm. It is used to optimize the UAV trajectories (Liu et al., 2021b). Indeed, it explores the topological information of any environment, such as obstacles. The algorithm tries to avoid them as much as possible while searching the shortest paths at each path point. For example, in Wang et al. (2020d), deployed multiple UAVs to perform DC over an IoT environment. VDM method is applied to divide the environment into multiple subregions, each of which is served by a given UAV data collector. The idea behind this technique is to reduce the DC completion time by jointly optimizing the UAV trajectory and transmit power of IoT devices.

Probabilistic RoadMap (PRM)

It is a path-planning algorithm that operates in two phases: (i) construction phase and (ii) query phase (Fu et al., 2022b). The construction phase converts the environment into an undirected graph. Then, the shortest paths are calculated based on probabilities during the query phase. For instance, the authors of Pěnička et al. (2019) deployed a UAV-based DC system mainly focusing on maximizing the collected data under the constraint of obstacles and limited energy of the deployed UAV. PRM combined with variable neighborhood search (VNS) algorithm to build the shortest collision-free trajectory and extend the initial roadmap.

Rapid random tree (RRT)

It is a path-planning algorithm proposed as an alternative to PRM to handle situations that consider non-holonomic and dynamic constraints frequently met in UAV environments (Guo et al., 2022). For instance, Asadi et al. (2020) proposed a novel UAV-UGV-based system for data collection in construction sites. RTT algorithm is adopted to establish a collision-free path for both UAV and UGV. This system is evaluated in a realistic environment, and the generated performances demonstrate the solution's effectiveness.

Artificial potential field (APF)

It is a path-planning algorithm producing attractive and repulsive forces from the different points of the environment (Jayaweera and Hanoun, 2020). The points that generate attractive force actively drag the UAV towards them. However, the points that generate repulsive force, the UAV is repelled from them. In that way, the UAV will find the path to its destination without colliding with obstacles. For instance, a path-planning collision-free approach in the context of UAV-based DC is proposed in Tazibt et al. (2018). Indeed, APF allows the UAV data collector to avoid obstacles by representing obstacles as repulsive forces and each IoT device as an attractive force. The objective is to increase the QCD while minimizing the energy consumption of the UAV.

Lessons learned

Our comprehensive study shows that the movement control of UAVs is not adequately optimized to efficiently overcome the restricted energy of both UAVs and IoT devices/sensors, as distinguished in Goudarzi et al. (2019) and Zhan and Zeng (2019b). From Li et al. (2020c), it can also be learned that applying continuous actions to UAVs may not correctly cover ground devices. Thus, the DC process cannot increase the quantity of collected data. Moreover, when transmitting their data towards UAVs, ground devices may cause interferences corrupting generated data from devices, which is the worst problem impacting real-time applications. These reviews allowed us to summarize the following recommendations to avoid such issues as follows:

- The trajectories of UAVs and their energy consumption should be jointly optimized.
- The energy levels of ground devices should be checked before any DC process to avoid their failures and, thus, lost packets.
- The generation of data at the ground level should be reliably managed using, e.g., efficient MAC protocols to avoid interferences.
- Continuous actions should be avoided to allow UAVs adequately cover ground devices and serve as many ground devices.

4.2. Data transfer in UAV-based data collection

The UAV-based DC process requires efficient data transfer schemes to ensure the reliability of data transmission towards their final destinations (Chenthil and Jesu Jayarin, 2022; Gurewitz et al., 2022). Several methods could be considered to reduce packet losses when data are routed between sensors or UAVs, such as avoiding collisions when sensors transmit their data towards UAVs, relaying data through UAVs, and transporting data through UAVs (c.f., Fig. 7). For this purpose, several schemes have been proposed, which are categorized into three classes: (i) Routing, (ii) Store-and-Forward, and (iii) MAC, which are covered in this section and summarized in Table 8.

4.2.1. Routing

Routing data in a multi-hop fashion is often used in the DC process, particularly when the destination is out of range of the source sensor (Haider et al., 2022). For instance, an energy-efficient routing protocol for UAV-assisted WSN is proposed in Baek et al. (2019). This work aims to maximize the residual energy levels of sensors after transmitting data to the UAV that plays the role of a traveling data sink. An optimization problem was derived to define the shortest routing path while considering sensor energy status and adjusting each UAV hovering location. Say et al. (2016) designed a UAV-based DC over WSNs, where the UAV also acts as a relay. To optimize the DC process's efficiency and minimize sensors' energy consumption, a routing protocol and priority access are designed to allow sensors to transmit their data efficiently while reducing their packet losses. In Okcu and Soyuturk (2014), a hybrid and energy-efficient clustering approach is proposed, where a UAV is deployed as a mobile sink collecting data from a WSN. The approach consists of selecting the sensor with the highest energy level and the nearest position to the UAV as a (CH) to which all other sensors are connected and communicate via multiple hops (i.e., a routing protocol). Authors in Ho et al. (2015) have proposed a Particle Swarm Optimization (PSO)-based algorithm for establishing clusters of sensors. This scheme aims to find the optimal topology to increase the lifetime of the whole system, maximize the QoS of communication, and reduce the traveling time of the UAV data collector. Ebrahimi et al. (2018) proposed a UAV-based DC strategy that leverages SCA method with the objective of decreasing both the sensors' energy consumption of sensors and the UAV flight time. Furthermore, the clustering formation, the election of CHs, the UAV trajectory, and the route paths in each cluster are jointly optimized.

4.2.2. Store-and-Forward (SaF)

When there is a single UAV data collector or multiple UAVs disconnected from each other, carrying data for a certain distance becomes unavoidable (Wei et al., 2018). This technique is well-known as Store-and-Forward (SaF), mainly applied by delay-tolerant applications (Lakew et al., 2021). For instance, a multi-UAV-based DC scheme is introduced in Rozaliya et al. (2022) to collect data from sensing locations in a SaF manner. The UAVs cooperate as relays to transfer data to maximize collected data at the BS while adhering to idleness and latency restrictions. Liu et al. (2018c) proposed a UAV-based DC SaF mechanism to collect data from all sensors and deliver them to the BS when it comes back to it. The trajectory of the UAV is optimized using a greedy algorithm in order to ensure a certain freshness of the gathered information. The authors of Abdulla et al. (2014) deployed a UAV data collector, flying in a circular trajectory to collect data from a set of sensor clusters. Data transmission scheduling is adopted to ensure DC fairness while increasing the energy efficiency of sensors. In Jawhar et al. (2015), a UAV-based delay-tolerant DC mechanism is proposed to collect data from a clustered WSN where the UAV's mission is to efficiently visit each CH. The SaF is adopted to collect non-critical data aggregated at each CH to be made accessible to the visiting UAV. Meanwhile, the authors in Ahmed et al. (2020) maximized the energy efficiency of UAV-based DC by jointly optimizing the trajectory and speed of UAV, the transmission power of both BS and UAV, and UAV propulsion energy consumption. This optimization allows the UAV data collector to efficiently relay and transport data to their intended destinations.

4.2.3. Medium access control (MAC)

When sensors transmit their data simultaneously, there is a risk of causing data collisions (Patel et al., 2021). Adopting channel access schemes could be the appropriate instrument to cope with this issue (Sun et al., 2021). The authors in Araghizadeh et al. (2016) considered a UAV-based DC using a priority MAC scheme to be deployed for emergency and monitoring applications. This scheme operates in four steps: (i) The UAV sends a beacon to all sensors within its LoS,

(ii) sensor receivers that have not yet registered for a transmission slot attempt to do so, (iii) The UAV creates a Time Division Multiple Access (TDMA) schedule and delivers it to every sensor that has signed up, and (iv) The sensor transmission process in TDMA-based slots is repeated. Mori (2016) proposed a cooperative communication using a UAV data collector for WSNs based on CSMA/CA MAC protocol. Indeed, each sensor has three operating modes: listen, transmit, and receive. To allow a sensor to transmit data, its mode has to be changed to transmit; otherwise, its mode should be changed to receive where it overhears the data transmission of other sensors. The mode of each sensor is reset to listen after each data transmission. Similarly, the study in Sotheara et al. (2014) also considered a CSMA/CA protocol, which is associated a priority-based frame selection scheme to improve the UAV-based DC QoS. This study adjusted the contention window to handle transmission priorities. Zanjie et al. (2014) attempted to maximize the data transmission rate through UAV data collectors. Indeed, the authors developed an optimal dynamic programming-based algorithm (DPBA) operating at the MAC layer, in which the bandwidth, UAV position, and energy allocation are jointly considered. In Vlăduță et al. (2016), a new MAC protocol for UAV-based DC is proposed to provide an equal opportunity for each sensor to communicate with the UAV data collector. This protocol operates based on three phases: (i) initialization, (ii) prioritization, and (iii) transmission. The first phase allows the UAV to make its presence known for all sensors, the second phase allows sensors to avoid transmission collisions, and the third phase allows sensors to make their data transmission towards the UAV. In Ruan et al. (2020), a novel MAC protocol is proposed allowing UAV data collectors to transmit collected data to the CC through the HAP relay. This protocol aims to reduce end-to-end connectivity while maximizing throughput.

Lessons learned

At the transmission level, the UAV-based DC process could be targeted by various problems before, during, or after the DC. These problems might occur due to potential vulnerabilities distinguished when uploading information to the UAV or transporting it. Indeed, as in Say et al. (2016), some proposed routing protocols did not consider the energy consumption of ground nodes and interferences that may occur between them when relaying information to the CHs. Furthermore, overhearing and idle listening are vast wastes of energy. Finally, UAV data collectors' trajectory and speed are not optimal enough to transport information to their intended destinations. Therefore, the following precautions should be given serious consideration:

- Energy-efficient interference-aware routing protocols are suggested to overcome packet losses and induced delays while uploading information through relay ground nodes.
- Reliable selection schemes are recommended, which should carefully define cooperative target nodes and the data to be retransmitted.
- Synchronization schemes need to be considered to reduce collisions and overcome the problem of interference.

4.3. UAV-based time-sensitive data collection

For the right functionality of certain applications, UAV data collectors need to continuously capture fresh information from IoT devices or sensors (c.f., Fig. 8). This DC process should consider the energy consumption of both UAVs and ground devices, the scheduling of data generation, and balancing between the device's uploading time and UAV flight time (Liang et al., 2022). To measure the freshness of gathered data, the AoI metric is used, which is already defined in Section 2.3.4. In Hu et al. (2020), two heuristic algorithms were adopted to minimize the average AoI of collecting data while jointly optimizing the UAV's path and time needed for DC and energy harvesting at each sensor. With a game theory viewpoint, Wang et al. (2022f)

Table 8

Summary of contributions of data transfer in UAV-based data collection.

		Objective	Advantage	Drawback	Cm	In	El	Mob	Se	Tsd	La	De
Routing	Baek et al. (2019)	Decreasing the energy consumption of sensors by optimizing the placement of the UAV data collector.	Operating better in dense WSN.	It did not consider the UAV data collector's energy consumption and is not suitable for real-time applications.	√	×	√	×	×	×	×	×
	Say et al. (2016)	Increasing the lifetime of sensors and the efficiency of the DC process.	An energy-efficient routing for maximizing the system throughput and reducing packet losses.	The placement of the UAV data collector is estimated without considering its energy consumption and interferences when CHs transmit their data simultaneously.	×	×	√	×	×	√	×	×
	Okcu and Soyuturk (2014)	Constructing well-based and energy-efficient clusters allows the UAV data collector to cover the maximum number of sensors.	Preventing unnecessary cluster formation using the UAV data collector route as a parameter to establish the clusters.	The UAV travel time over a large-scale sensor environment is not considered.	×	√	√	√	×	×	×	×
	Ho et al. (2015)	Decreasing the energy consumption of sensors while reducing the traveling time of the UAV data collector.	Improving the overall system lifetime in terms of energy consumption.	The UAV energy consumption is not accurately estimated as it can be affected by the quantity of data to be collected.	√	√	√	×	×	×	√	×
	Ebrahimi et al. (2018)	Saving energy of sensors and extending WSN lifetime.	Easily adapted to dense WSNs.	Increasing the flight time of the UAV data collector thus consumes more energy.	×	×	√	√	×	×	√	×
Store and Forward	Rozaliya et al. (2022)	Maximizing collected data received at the BS while satisfying the latency and idleness constraints.	The cooperation of UAVs has proven its efficiency in the amount of collected data.	La is not always guaranteed through the shortest path of UAVs, and dense WSNs require high computational time.	×	×	×	√	×	×	√	×
	Liu et al. (2018c)	Increasing the freshness of collected data.	Efficiently finding the age-optimal trajectories of UAVs.	Did not consider the scheduling of packet transmissions when optimizing the UAV trajectory.	√	×	×	√	×	√	×	×
	Abdulla et al. (2014)	Increasing the energy efficiency equitably, <i>i.e.</i> , the data rate per energy.	Enhancing the energy efficiency of the whole system	The movement of CHs could impact the UAV movement and thus negatively influence the quality of data transmission.	√	×	√	×	×	×	×	×
	Jawhar et al. (2015)	Reducing interference and energy consumption in the data transmission and extending the system lifetime.	A real-time data transfer model assists SaF in increasing the freshness of collected data.	This hybrid model is inadequate for real-time applications, requiring time-sensitive data for proper functionalities.	×	√	√	×	×	×	√	×
	Ahmed et al. (2020)	Validating the performance of a proposed energy-efficient UAV trajectory for a DC process.	Decreasing the UAV energy consumption both in the movement and communication.	Overlooking interferences and considering only a single sensor and a single UAV-assisted relay.	√	×	√	√	×	×	×	×
MAC	Araghizadeh et al. (2016)	Providing high throughput in dense WSN while guaranteeing fairness among sensors.	Increasing throughput and fairness by about 20% and 25%, respectively.	Did not consider the mobility of the UAV and its energy consumption.	√	×	×	×	×	√	√	×
	Mori (2016)	Cooperative communications using an improved MAC protocol to increase the efficiency of the DC process.	Sensors significantly minimize frame loss by receiving, storing, and transmitting overheard data.	Taking time and energy to determine which sensing data should be retransmitted.	√	×	×	√	×	×	×	×
	Sotheara et al. (2014)	Allowing data with higher priority to be transmitted first.	Achieving better performances in terms of throughput, delay, and delivery ratio, while significantly reducing the packet losses.	The UAV data collector is assumed to be hovering, which neglects the realistic deployment of such systems.	√	×	×	×	×	√	√	×
	Zanjie et al. (2014)	Maximizing the data rate of sensors while ensuring a stable data rate at each time-slot for every sensor.	Allocating bandwidth and energy increases the total transmission data rate for all sensors.	Flight is significantly increased due to the hovering of UAVs. Moreover, interferences could occur since multiple UAVs are carrying out DC.	√	×	√	×	×	×	×	×
	Vlăduță et al. (2016)	Providing a fair chance for sensors to transmit their data to the UAV data collector.	Reducing collisions and packet losses in the network.	High congestion is distinguished due to control packets.	√	×	√	√	×	×	×	×
	Ruan et al. (2020)	Allowing multiple UAVs to communicate with the HAP using a dynamic channel allocator parameter according to the density of UAVs.	Achieving high throughput and less end-to-end delay than traditional MAC protocols.	Did not consider the negative impact of residual time deviation after synchronization.	×	×	×	√	×	×	×	×

optimized the AoI in the context of UAV-aided MEC networks by considering channel access attacks during the DC process. The authors in Jia et al. (2019) employed the dynamic programming (DP) technique to minimize the overall AoI by jointly optimizing the UAV's flying and hovering and the energy consumption of IoT devices. To minimize

the maximum AoI while balancing the UAV's flight time and the IoT device's uploading time, Tong et al. (2019) adopted a DP technique to optimize both the UAV's trajectory and the sensors-collection points association. In Wu et al. (2021c), an RL strategy is used to ensure a certain freshness of collected data while minimizing the energy consumption

Table 9

Summary of contributions of UAV-based time-sensitive data collection.

	Objective	Advantage	Drawback	Cm	In	El	Mob	Se	Tsd	La	De
Time sensitive data collection	Hu et al. (2020)	Minimizing the AoI of the system while jointly considering the UAV trajectory and the time needed for energy harvesting and DC.	The minimal average AoI of the system is obtained, and the near-optimal average AoI is founded.	Working only with a single UAV scenario while neglecting its energy consumption.	✓	✓	✓	✓	×	✓	×
	Wang et al. (2022f)	Minimizing the AoI while considering channel access attacks and the active probability of the system.	Achieving better AoI performance under the constraints of security issues.	The throughput of ground devices is not considered to minimize the AoI, especially when updates have different sizes.	✓	✓	×	×	×	✓	×
	Jia et al. (2019)	Minimizing the AoI while jointly optimizing the UAV trajectory, DC mode selection, and energy levels of ground devices.	Obtaining an optimal flight time strategy to reduce the AoI.	Did not consider both large-scale IoT environments and UAV energy consumption.	✓	×	✓	✓	×	✓	×
	Tong et al. (2019)	Minimizing the maximum AoI of all sensors while balancing between the UAV's flight time and sensors' uploading time.	Obtained a better performance in terms of AoI by optimizing the UAV trajectory.	The UAV failure scenario is not considered, and the authors did not show how to handle this situation.	✓	×	×	✓	×	✓	×
	Wu et al. (2021c)	Minimizing the AoI of multiple UAVs by jointly determining the path during sensing and transmission.	Obtaining a lower average of AoI compared to baseline methods.	Collision avoidance is not considered, and the ground devices are supposed with a sufficient energy supply.	✓	✓	✓	✓	×	✓	×
	Abd-Elmagid et al. (2019)	Achieving the minimum weighted sum AoI while optimizing the UAV trajectory and the updates' scheduling sensors.	Maintain a certain level of collected information's freshness.	The UAV energy consumption is not considered.	✓	×	✓	✓	×	✓	×
	Mao et al. (2020)	Minimizing the AoI of large-scale sensor environments by deploying multiple energy-constrained UAVs.	Brought more gain in terms of AoI performance compared to baseline methods.	Collision avoidance and UAV failure scenario are not considered.	✓	×	✓	✓	×	✓	×
	Liu et al. (2020e)	Striking a balance between the UAV flight time and sensors' uploading time.	Enhancing the freshness of collected information from all sensors.	The UAV energy consumption is not considered.	✓	×	×	✓	×	✓	×
	Wu et al. (2021a)	Deploying an AI-based UAV trajectory to minimize the AoI.	Determining the optimal hover positions of the UAV that increases the efficiency by 200%.	The UAV energy consumption is not considered, which cannot allow us to consider UAV's failure scenarios.	✓	×	✓	✓	×	✓	×
	Ghdiri et al. (2021)	A set of heuristic algorithms to reduce AoI of collected data from time-constrained ground devices.	Increasing the amount of collected data without prior knowledge of the IoT environment.	No collision avoidance mechanism.	✓	×	✓	✓	×	✓	×

of the UAV data collector by optimizing its trajectory. Similarly, the authors in [Abd-Elmagid et al. \(2019\)](#) used an RL strategy for both UAV trajectory optimization and scheduling the sensors' updates. The purpose of this strategy is to achieve the minimum weighted sum of AoI. To reduce the AoI of collected data, [Mao et al. \(2020\)](#) optimized the trajectories of multiple UAVs using a DP technique and k-means algorithm to locate the positions of the downloading points. In a similar solution ([Liu et al., 2020e](#)), the authors optimized the trajectory of the UAV data collector in an energy-efficient manner to balance the UAV flying time with the sensors' uploading time under different conditions. In [Wu et al. \(2021a\)](#), the authors presented an end-to-end AI-based system to handle the UAV trajectory planning in two stages, striking a good compromise between accuracy and efficiency. First, a clustering module is used to determine the UAV's hover locations and the duration of the DC process. Then, a neural trajectory solver is used to determine the AoI-minimal flying route. To maximize the amount of freshly collected data from time-constrained IoT devices, the authors of [Ghdiri et al. \(2021\)](#) adopted a combination of algorithms (*i.e.*, Tabu-search, simulated annealing, guided local search, and RL) to optimize various metrics of multiple UAVs in the offline and online settings. Moreover, they used the K-means method to group sensors and select CHs. All the time-sensitive contributions discussed in this subsection are summarized in [Table 9](#).

Lessons learned

When the UAV-based DC concept is deployed to support real-time applications, its proper functioning strongly depends on the freshness of

collected information. However, as distinguished in [Wang et al. \(2022f\)](#) and [Jia et al. \(2019\)](#), it is not easy to guarantee this requirement without considering key challenges, such as the throughput of ground devices, the coverage optimization, the energy consumption of all energy-constrained nodes, and the trajectories of UAV data collectors. First, some generated data packets have different sizes, which require higher throughputs to be uploaded quickly to reduce the AoI metric. Second, according to the application, some scenarios are characterized by their large-scale IoT environments that need to be optimally and timely served, which is impossible using a single UAV or a reduced number of UAVs. Finally, the energy consumption of UAVs and ground devices should be considered during their movements and communications to avoid node failure and, thus, a severe increase in the AoI. Therefore, there is a need to study and apply some solutions to reduce the AoI metric. In the following, we name a few:

- Optimizing both the transmission power of devices and the altitude of UAVs increases the throughput and thus reduce upload delays.
- Deploying the appropriate density of UAVs to optimally and timely serve ground environments.
- Reducing the energy consumption of ground devices and UAVs by optimizing the UAV trajectories and the data generation at the level of ground devices.

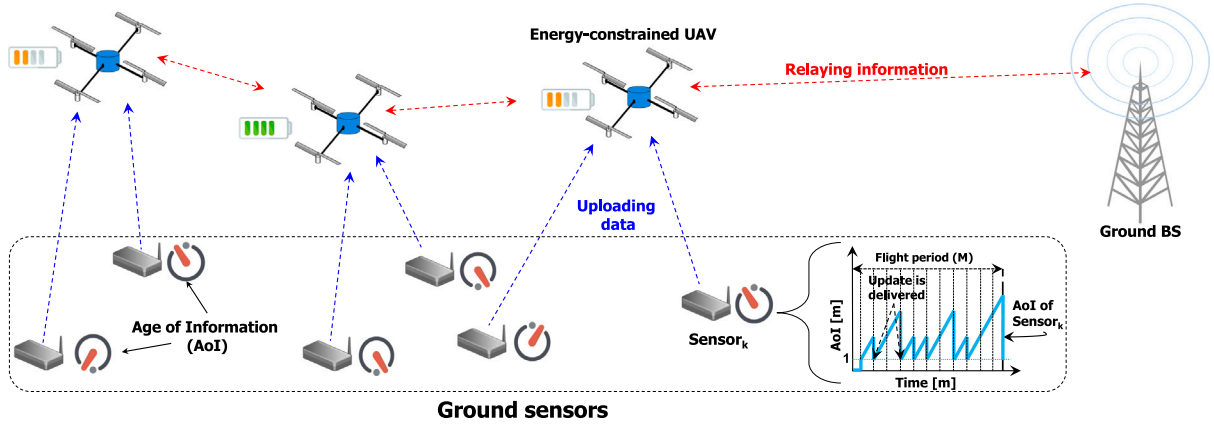


Fig. 8. UAV-based time-sensitive DC over sensors.

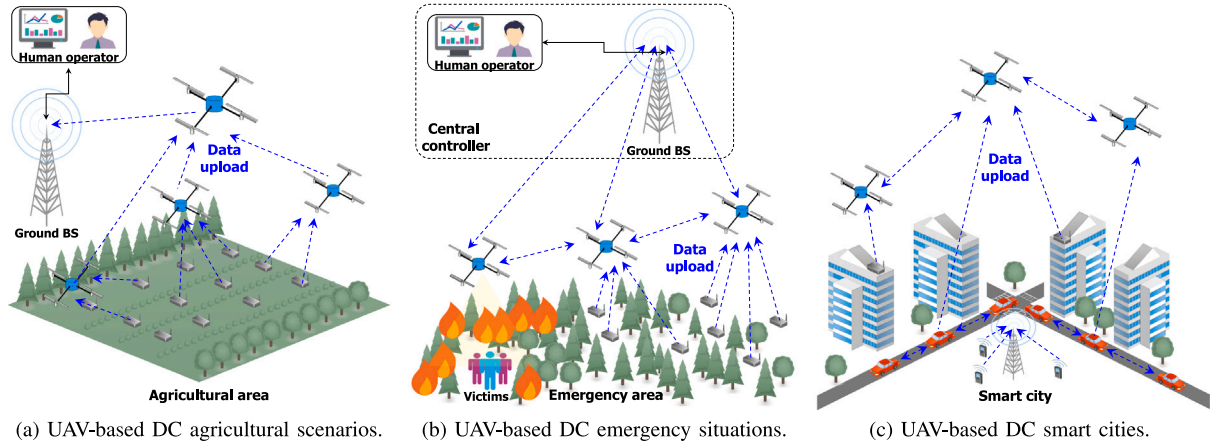


Fig. 9. UAV-based DC applications.

4.4. UAV-based data collection applications

Several applications adopt UAVs to perform various tasks to explore different environments, assist ground users and IoT devices, and relay gathered information to the CC for processing (Alsamhi et al., 2021). In the literature, we distinguished three main categories of UAV-based DC applications: (i) Agricultural, (ii) Emergencies, and (iii) Smart cities, which are covered in this section and briefly summarized in Table 10.

4.4.1. Agricultural

In agricultural fields, farmers need to have accurate and timely details about their crops, which are obtained through the soil and air measurements and then used for agricultural operation management (e.g., irrigation control, pesticide doses, and pest management) (Ojha et al., 2021). For this purpose, UAVs and IoT devices are integrated into agricultural fields to perform data gathering and processing (Madrikunta et al., 2021). This architecture can assist farmers in defining irrigation plans and the amount of fertilizer needed and thus enhancing the agricultural goods while saving money on fertilizers and pesticides by using a data-driven strategy (c.f., Fig. 9(a)). For instance, Wang et al. (2018) proposed a UAV-based crop information collection to monitor the crop growth status. This monitoring is based on sensors placed on the crop field. Similarly, the authors in Popescu et al. (2020) proposed a cooperative hierarchical system for crop health monitoring. This system is a collaboration method of UAV and WSN where the UAV's trajectory is optimized, and relevant data transmission is assessed through effective data processing algorithms. In Singh and Sharma (2022), the authors deployed fixed-wing UAV and WSN architecture for spraying and crop monitoring tasks where the UAV trajectory and

data analysis are jointly optimized. Tsourous et al. (2019) discussed agriculture picture collection schemes using a UAV for monitoring crop field in precision agriculture. In Zhou et al. (2021), a UAV sleep schedule ensures sufficient coverage with the least number of sensors deployed on a crop field. Moreover, The UAV placement is optimized through the Monarch Butterfly Optimization (MBO) algorithm. The authors of Andreadis et al. (2022) deployed a set of HAPs over an IoT environment deployed in agricultural areas for monitoring ground data and smart agriculture. This architecture uses the LoRa/LoRaWAN technology to allow IoT devices to transmit their sensed data over long distances.

4.4.2. Emergency situations

In disaster scenarios, there are three stages where UAVs could constitute essential support to overcome these scenarios with the least possible damage (Qadir et al., 2021). First, UAVs could reach difficult locations and perform real-time monitoring to identify disasters. Second, UAVs could act as rescue members and participate in seeking missing victims and providing help. Finally, UAVs could cooperatively establish a connected communication infrastructure to allow terrestrial rescue members to communicate with the CC. In all these stages, the UAV-based DC process is ubiquitous and helpful in efficiently performing each task (c.f., Fig. 9(b)). Poudel and Moh (2021) optimized the trajectories of multiple UAVs to efficiently collect data and avoid collisions between them in disaster situations. For this purpose, the authors used a hybrid path planning (HPP) algorithm, which is a combination of the probabilistic roadmap (PRM) and optimized artificial bee colony (ABC) algorithms. The authors in Duong et al. (2019) adopted relay-assisted UAVs in WSN in natural disaster scenarios where

Table 10
Summary of contributions of UAV-based data collection applications.

		Objective	Advantage	Drawback	Cm	In	El	Mob	Se	Tsd	La	De
Agricultural	Wang et al. (2018)	Managing farmland crop information gathered by UAVs.	Visualizing collected data in real-time.	Did not take into account the UAV's energy usage and the freshness of collected data.	×	×	×	√	×	×	×	×
	Popescu et al. (2020)	UAV-WSN collaboration for crop monitoring.	Increasing the performances in terms of intelligent farming and ecological agriculture.	The gathered data is not combined to produce a unified output.	×	×	√	√	×	×	×	×
	Singh and Sharma (2022)	Enhancing the data extraction and better synchronizing between UAVs and WSNs for data gathering.	Increasing the coverage efficiency of 96.3% in different agricultural applications, such as crop monitoring or spraying pesticides.	Did not consider the computing efficiency during the data extraction and the efficiency of UAV-WSN communication.	×	×	√	√	×	×	√	×
	Tsouros et al. (2019)	Summarizing technologies for image acquisition in the context of UAV-based precision agriculture and farming operations.	Efficiently detecting weak crops field and making timely decisions.	Required intelligent methods to correctly avoid false positives.	×	×	×	×	×	√	×	×
	Zhou et al. (2021)	Ensuring sufficient coverage with the least number of sensors while avoiding excessive data exchange with them.	Extending the network lifetime while guaranteeing the appropriate coverage.	Dedicated only to a single UAV scenario and did not repair network holes.	×	×	√	×	×	×	×	×
	Andreadis et al. (2022)	Allowing IoT devices to transmit their captured data through the LoRa/LoRaWAN technology to the internet through HAPs.	Achieving a high probability of transmitting data from sensors for long distances.	The restricted energy of IoT devices is not considered.	×	×	×	√	×	×	×	√
Emergency Situations	Poudel and Moh (2021)	Ensuring an efficient UAV-based DC through the shortest collision-free paths in emergency environments.	Enhancing the designed system in terms of energy consumption, UAV flight time, and convergence time.	It did not improve coverage in large-scale scenarios and is not adapted to dynamic obstacles.	×	×	√	√	×	×	√	×
	Duong et al. (2019)	Optimizing the trajectories of relay-assisted UAVs over WSNs to overcome the hazardous effect of natural disasters.	A low execution time and computational complexity demonstrate the benefits of the proposed optimization method for UAV-WSN collaboration.	The UAV and WSN energy consumption is not considered.	√	×	×	√	×	×	×	×
	Erdelj et al. (2017)	Managing WSN-UAV in natural disaster scenarios.	Enhancing WSN and multi-UAV collaboration in different application domains in disaster management while overcoming different distinguished challenges.	No standard measure is taken against the restricted energy of sensors and UAVs.	×	×	×	√	√	×	×	×
	Wang et al. (2020a)	Achieving a trade-off between the energy cost and the DC ratio.	Increasing considerably the DC ratio and reducing the energy cost.	It is not adapted to large-scale scenarios with an essential number of ground nodes.	×	×	√	×	×	×	×	×
	Demiane et al. (2020)	Accurately localizing an undetermined number of devices in disaster areas.	Demonstrating the ability of the solution to adapt to regions with varying importance levels.	It is not adapted to large-scale scenarios.	×	×	×	√	×	×	×	×
Smart Cities	Shen et al. (2021a)	Ensuring the credibility of UAV-based DC by identifying the credibility of sensors.	Quickly identifying the trust of sensors while reducing the flight path of UAVs.	The sensors' energy consumption is not considered.	×	×	√	√	√	×	×	×
	Zhang et al. (2017)	Intelligently controlling UAVs and mobile charging stations using RL.	Obtaining the highest DC rate compared to other baseline methods.	Did not consider the energy consumption of the mobile charging station.	×	×	√	√	×	×	×	×
	Huang et al. (2021a)	Ensuring a secure UAV-based DC through an active trust verification scheme.	Achieving high efficiency in verifying the reliability of mobile vehicles.	Focusing only on data and neglecting the interaction between nodes.	×	×	√	√	√	×	×	√
	Lee et al. (2021)	Collecting monitoring data while considering the energy constraint of UAVs using distance and data similarity.	Achieving the desired performance in terms of collected data and UAV energy consumption.	Requires a significant convergence period.	×	×	×	√	×	×	×	×
	Tan et al. (2019)	Collecting delay-sensitive information while optimizing the energy efficiency of the system.	Demonstrating 20% improvement in terms of energy efficiency compared to benchmark methods.	It did not consider a possible UAV failure and how to deal with it.	√	√	√	×	×	√	√	×

low-complexity optimization techniques are adopted to deal with a restricted execution time in such scenarios. To perform a real-time assessment of a flood disaster, Erdelj et al. (2017) combined UAVs and WSNs to carry out detection, size evaluation, and localization of flooded areas through UAV-derived aerial images. In Wang et al. (2020a), the authors proposed an intelligent DC mechanism based on a task distribution mechanism that ensures a trade-off between the DC

ratio and energy consumption. Moreover, a RL strategy is adopted to optimize the UAV trajectory and ensure the relaying of collected data through a connected UAV network. Demiane et al. (2020) optimized the UAV trajectory, which is deployed for searching for survivors and tracking terrestrial mobile targets in disaster scenarios while using the DC concept to perform these tasks.

4.4.3. Smart cities

A set of smart devices are deployed in smart cities to improve the quality of life and services of their inhabitants (Al Ridhawi et al., 2021). Many applications and services are mainly based on the use of UAVs and IoT devices where the DC process constitutes an integral part of ensuring their proper functionalities (c.f., Fig. 9(c)). For instance, Shen et al. (2021a) proposed a secure UAV-based DC approach combining UAV tracking back and digital signature mechanisms. This approach aims to detect trusted and malicious nodes on the ground to secure the DC process. The authors in Zhang et al. (2017) applied an RL strategy to optimize the UAV and mobile charging station trajectories and enhance the DC process in smart cities, which considerably increases the mission's duration and efficiency. Always in a smart city scenario, in Huang et al. (2021a), a UAV data collector is deployed to collect baseline data from sensors embedded in moving vehicles in a smart city. The purpose of this scheme is to achieve DC while maintaining an acceptable level of safety, efficiency, and UAV flight cost. Lee et al. (2021) proposed a UAV-based DC approach that selectively collects data from CCTV cameras under the consideration of data similarity and distance instead of gathering data from static points of interest. The purpose of this approach is to maximize the seller's revenue using a deep learning auction method. In Tan et al. (2019), a UAV-based DC mechanism is proposed to collect time-sensitive data and reduce energy consumption in a smart public place. For this purpose, the transmission power of IoT devices is adjusted, and the UAV 3D placement is optimized based on a heuristic approach.

Lessons learned

Recently, there has been extensive involvement of the UAV-based DC concept in various agricultural, emergency, and smart city applications. However, through our study in different applications (Tsouros et al., 2019; Duong et al., 2019; Huang et al., 2021a), we distinguished a set of unaddressed drawbacks that require careful attention, such as the nodes' failures due to lack of energy, neglecting the interaction between nodes, and considering false positives. Indeed, both UAVs and ground devices could suffer anytime from a lack of energy, and their failures could disturb the whole application. Moreover, the communication between nodes is generally overlooked, which could expose the system to having an unclear vision of the environment and thus gathering false positive information. To address these issues, we recommend some precautionary measures outlined as follows:

- The energy status of each energy-constrained should be permanently updated to take the necessary measures in terms of charging or battery replacement and thus avoid distributing the whole application.
- Leveraging the communications between nodes to have an accurate vision of the environment and avoiding false positive information gathering.
- Deploying the appropriate density of UAVs to ensure the proper functionality of the application.

4.5. Energy efficient UAV-based data collection

Based on energy-efficient techniques, the DC process can significantly decrease the energy consumption of IoT devices/sensors and UAVs (c.f., Fig. 10). For example, a transmission/wakeup scheduling optimization or CH election can be used to allow IoT devices or sensors to transmit their data with fewer energy constraints (Zhu et al., 2022b; Sheshashayee et al., 2022). Moreover, the trajectories of UAV data collectors are also optimized to reduce the traveled distance, and thus the consumed energy (Li et al., 2022a). Wang et al. (2019b) optimized the transmission/wakeup scheduling to maximize the sensors' energy efficiency. Furthermore, the authors optimized the UAV trajectory while guaranteeing the collected data amount from sensors. In Kim and Qiao (2020), the authors deployed multiple UAVs acting as relays between

remote IoT devices and a BS. An energy-efficient and cooperative multi-hop relay strategy is proposed to minimize the energy consumption of UAVs while considering their given time constraint. The authors in Qin et al. (2019) deployed multiple UAVs with different energy capacities to maximize the quantity of collected data by jointly optimizing the UAVs' trajectories and communication scheduling, thus reducing their mission completion time. In Wang et al. (2020d), the authors deployed multiple data collectors in a hovering or flying mode to reduce energy consumption and maximize the amount of collected data. Zhan et al. (2017) jointly optimized the UAV trajectory and the wake-up schedule of sensors to reduce the energy usage of all sensors while ensuring that each sensor transmits the required quantity of data to the collector. The authors applied the successive convex optimization technique to solve this optimization problem. To reduce the UAV energy consumption and maximize the throughput in the UAV-assisted machine-type communications networks, Zhu et al. (2018) optimized the hovering positions of the UAV data collector using a heuristic algorithm. In Nazib and Moh (2021), the authors used a two-step scheme for improving the UAV-based DC. Firstly, a clustering method is adopted to find the sub-optimal downloading points' positions. Secondly, a genetic algorithm is adopted to optimize the trajectories of the UAV data collector. Ghdiri et al. (2020) optimized the required density of UAVs to be deployed as data collectors and their trajectories using a heuristic approach. Moreover, they also used the K-means clustering algorithm to reduce the overall energy consumption of sensors. The authors of Dong et al. (2019) deployed a fixed-wing UAV to perform DC and relaying of information. The continuous convex method and path discretization method are jointly used to optimize the trajectory of the UAV and reduce its energy consumption. In Wu et al. (2019), two modes of DC are used to collect data from sensors. In the first mode, a single UAV-based DC focuses on minimizing sensors' energy consumption while reducing the delivery delay of collected data. In the second mode, multiple UAVs-based DC, where UAVs are deployed over numerous sub-networks and cooperatively exchange data between UAVs. Jia et al. (2021) proposed a HAP-LEO-based data collection scheme deployed over remote devices. The primary purpose of this scheme is to maximize the quantity of collected data while considering the multiple resource constraints of HAPs and LEO satellites, such as their energy consumption and flow limitations. A summary of these schemes is given in Table 11.

Lessons learned

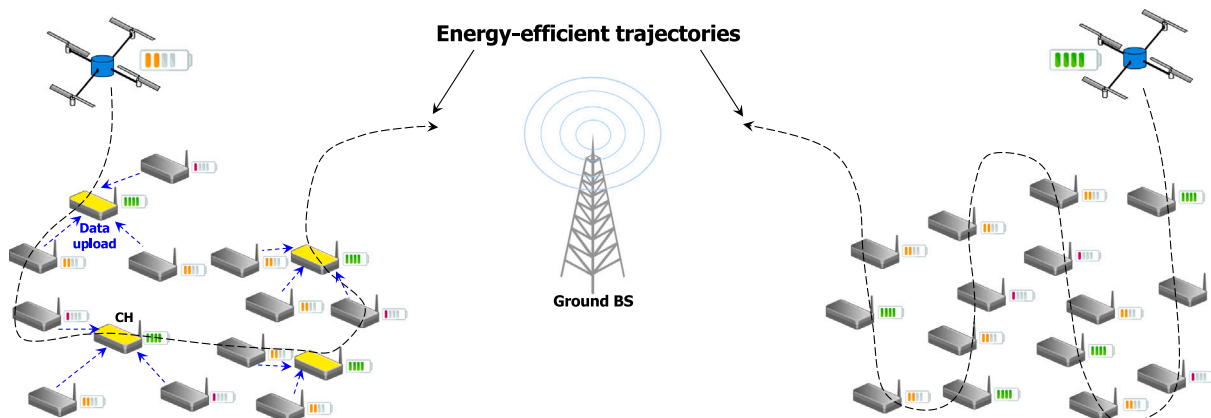
The practical development of UAV-based DC needs an emphasis on the energy efficiency principle. We learned through our reviews that some measures are taken to decrease the energy usage of UAV-based DC systems, which generates other issues related to networking communications, interferences, and overhead packets. For instance, in Kim and Qiao (2020), an energy-efficient multi-hop scheme is adopted, which does not consider the occasional disconnections disrupting remote transmissions. Moreover, some other schemes (Zhan et al., 2017; Ghdiri et al., 2020) involved a high number of overhead packets to periodically select CHs between ground nodes to upload data towards UAVs, which could cause the problem of interferences. Also, another method that could be adopted to increase the energy efficiency of a UAV-based DC system is energy harvesting. For instance, two interesting works in Agrawal et al. (2022b) and Agrawal et al. (2022a) combined several mechanisms, such as the incremental relaying along with the selective decode-and-forward relaying with the direct link and opportunistic cooperative mode to non-cooperative mode switching, to increase the performances of non-linear energy harvesting and thus decrease energy consumption. Therefore, there is a need to design the following countermeasures to avoid these issues:

- Selecting the most connected nodes having high energy levels to avoid disconnections due to occasional movements.
- Electing the most stable CHs with high energy levels and avoiding periodical re-elections to reduce overhead packets.
- Considering interferences between nodes by designing synchronization schemes to manage transmissions of nodes.

Table 11

Summary of contributions of energy efficient UAV-based data collection.

	Objective	Advantage	Drawback	Cm	In	El	Mob	Se	Tsd	La	De
Energy Efficient data collection	Wang et al. (2019b)	Decreasing the maximum power consumption of all IoT devices while ensuring DC reliability.	Brought more gain in terms of energy-efficient DC and 3D device position.	It did not consider the De induced by the UAV movement.	✓	×	✓	✓	×	×	×
	Kim and Qiao (2020)	Optimizing the multi-hop transmission scheduling to reduce the energy consumption of UAVs.	Efficiently relay the collected information while reducing the UAV energy consumption.	It did not address the UAV multi-hop network's link failures, which are unsuitable for remote transmission scenarios.	✓	✓	✓	✓	×	×	✓
	Qin et al. (2019)	Maximizing the amount of collected data by jointly optimizing the trajectories of UAVs and the transmission scheduling of sensors.	The DC utility is significantly increased based on the proposed approach.	Did not consider the 3D movements of UAVs to perform DC.	×	×	✓	✓	×	✓	✓
	Wang et al. (2020d)	Minimizing the flight time of UAVs while allowing energy-constrained devices to successfully transmit their data.	Maximizing the minimum throughput of devices while achieving a significant gain in DC performance.	No collision avoidance mechanism is considered.	✓	×	✓	✓	×	×	×
	Zhan et al. (2017)	Optimizing the UAV trajectory and sensors' wakeup to reduce the energy consumption of all sensors while guaranteeing that the required information is collected.	Achieving significant energy saving both in sensors and UAV.	Did not avoid interference between nodes.	✓	×	✓	✓	×	×	×
	Zhu et al. (2018)	Optimizing the UAV energy consumption and its throughput while increasing the efficiency of tasks and computing of IoT devices.	Algorithm with less complexity while minimizing the UAV energy consumption and throughput.	Did consider the energy constraints of IoT devices.	✓	×	✓	✓	×	×	✓
	Nazib and Moh (2021)	Reducing the energy consumption of sensors while achieving a fast DC.	Outperforming baseline methods in terms of scalability, the energy consumption of sensors, and load balancing.	Interferences are neglected, and the UAV energy consumption is not accurately computed.	×	×	✓	✓	×	×	✓
	Ghdiri et al. (2020)	Determining the best positions of a minimal density of CHs.	Increasing the efficiency of CHs placement while reducing the UAV energy consumption.	More complex when selecting CHs.	✓	×	✓	✓	×	×	×
	Dong et al. (2019)	Minimizing the propulsion energy of the fixed-wing UAV while ensuring a specific throughput requirement.	Obtaining the optimal UAV trajectory to efficiently collect and forward data.	Difficult to hover over IoT devices.	✓	×	✓	✓	×	×	×
	Wu et al. (2019)	Addressing the problem of enhancing the energy efficiency of WSN during the DC process.	Saving more energy in sensors and minimizing data delivery from sensors to the sink while reducing the UAV energy consumption.	Disability of DC due to the energy exhaustion of a CH, and thus it wastes the UAVs' residual energy.	×	×	✓	×	×	×	✓
	Jia et al. (2021)	Maximizing the QCD during a predefined period of time while considering the resource constraints of involved HAPs and LEO satellites.	The adopted optimization algorithm has proven its efficiency in increasing the QCD and reducing the energy consumption of HAPs and satellites.	No multiple access techniques are considered to allow multiple connections of HAPs to a given satellite.	✓	×	✓	×	×	×	×

**Fig. 10.** Energy-efficient UAV-based DC.

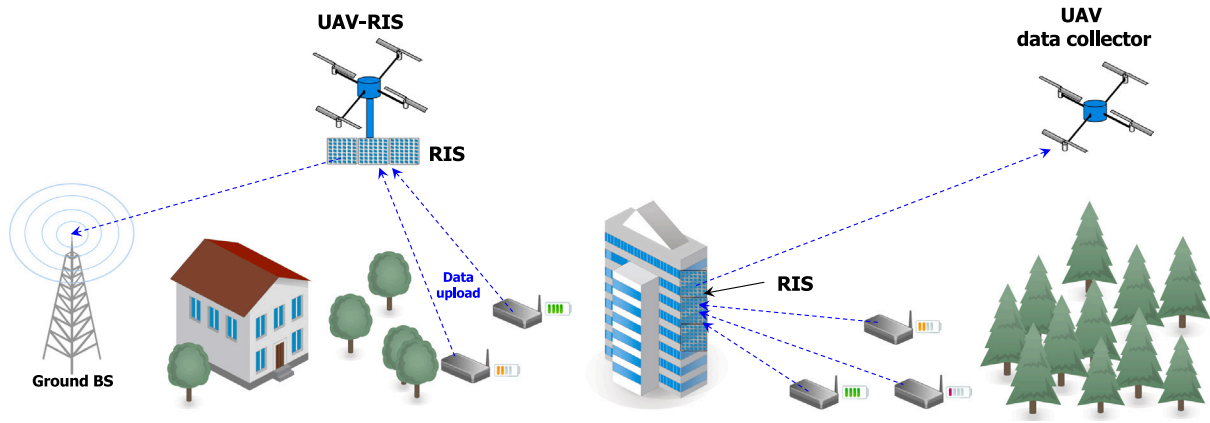


Fig. 11. UAV-RIS-based DC.

4.6. UAV-RIS-based data collection

A RIS is a device made up of many components that reflect Radio Frequency (RF) signals in the direction of the target receiver (Nguyen et al., 2022b). In the UAV-based DC scenarios, a RIS can be either mounted on the UAV or placed somewhere in the environment to enhance the communication quality between the communication nodes (c.f., Fig. 11). Therefore, our discussion in this category is based on two subcategories: (i) RIS-mounted UAV and (ii) RIS UAV cooperation.

4.6.1. RIS-mounted UAV

Embedding RIS in UAVs has also demonstrated its benefits in reducing the energy consumption of UAVs if UAV trajectories and resources are perfectly optimized (Liu et al., 2020b). For instance, the authors in Mahmoud et al. (2021) have deployed a UAV-mounted RIS to improve the data transmission efficiency by optimizing the spectral efficiency of IoT networks while assuming uncorrelated channels between transmitter/receiver and RIS. You et al. (2021) deployed UAV-mounted RIS-assisted IoT networks while considering the transmission power of IoT devices during the DC process. A speed-estimation-based channel tracking method is adopted to estimate the device's speed based on received signals and then predict the Angle of Arrival (AoA) and Angle of Departure (AoD) for designing the RIS passive beamforming. Nnamani et al. (2021) ensured a secure DC using a UAV-mounted RIS data collector where a non-iterative sub-optimal solution is adopted to maximize the secrecy rate. The authors jointly optimized the trajectory of the UAV-mounted RIS, RIS phase shifts, and sensors' beamforming weights.

4.6.2. RIS UAV cooperation

It is beneficial to adopt RISs since they improve the signal quality, decrease interferences, and prevent us from installing new stations (Basharat et al., 2021). For instance, in Ranjha and Kaddoum (2020), a UAV relay and RIS are deployed to facilitate an ultra-reliable low latency communications (URLLC) scenario. To address the overall decoding error probability minimization problem, the authors jointly optimized UAV trajectory, channel blocklength, and passive beamforming. Michailidis et al. (2021) considered multiple RISs installed on the walls to supply the communications between UAV and BS, which are acting as aerial and ground relays performing MEC functionalities where the time-slot scheduling, UAV energy constraints, and task allocation are jointly optimized. In Cao et al. (2021a), a RIS-assisted system is installed to support each UAV-device pair while optimizing the energy efficiency, RIS elements allocation, and their phase shift configuration. Furthermore, a multi-task learning strategy is adopted to minimize the complexity of the proposed transmission protocol. To increase the UAV-based DC efficiency, the authors of Al-Hilo et al. (2022) proposed an RL strategy to maximize the number of served

IoT devices while handling the configuration of the RIS using Block Coordinate Descent (BCD). To maximize the total network's sum rate, the authors in Nguyen et al. (2022c) employed a UAV and RIS to support IoT networks while jointly optimizing the phase shift matrix, UAV trajectory, and IoT energy harvesting schedule based on an RL strategy. Zhao et al. (2022) studied a scenario where multiple jammers are deployed to disturb the functionality of a UAV-RIS-based DC system. The authors aimed to maximize energy efficiency by jointly optimizing the UAV trajectory and the RIS's phase shift matrix while strengthening the desired signal and avoiding interferences.

A brief summary of these UAV-RIS-based contributions is presented in Table 12.

Lessons learned

RIS technology has become a center of interest for many researchers, especially when this technology involves UAVs. However, various issues are distinguished in some discussed references, such as in Ranjha and Kaddoum (2020) and Mahmoud et al. (2021) where the powering of RIS's elements is wholly overlooked, and interferences are not considered. Furthermore, when RIS devices are embedded in UAVs, we distinguished that the trajectories of UAVs are not optimized to avoid interferences and optimally serve large-scale IoT environments. Therefore, some solutions could be considered to avoid those issues as follows:

- Adopting models to estimate the energy consumption of RIS's devices and charge them.
- The trajectories of UAVs should be optimized to avoid interferences and serve the maximum possible ground devices.
- Reducing the AoI metric by optimizing the relay between multiple RISs.

4.7. MIMO for UAV-based data collection

MIMO technology can be described as a system equipped with multiple antennas to enhance the coverage capacity and throughput of a given wireless network (Maeng et al., 2022). When MIMO technology is used in UAV communications (c.f., Fig. 12), it can effectively solve the issues of the limited communication capacity and low transmission rate of UAVs by making full use of the channel capacity of the system without impacting the bandwidth and power (Bai et al., 2022). For instance, Hou et al. (2020) investigated a scenario where a full-duplex MIMO UAV-based DC system is deployed by considering the hardware impairments correlated with the input signal. The SCA method is adopted to optimize the UAV trajectory and beamforming transmission to minimize the energy consumption of UAVs. Alkama et al. (2022) deployed a UAV-based assisted network using 3D beamforming technology through the stochastic cellular network. The analysis of the

Table 12

Summary of contributions of UAV-RIS-based data collection.

		Objective	Advantage	Drawback	Cm	In	El	Mob	Se	Tsd	La	De
RIS-mounted UAV	Mahmoud et al. (2021)	Extending network coverage and improving spectral efficiency and communication reliability of IoT networks.	Increasing the performance in terms of symbol error rate using RIS.	Several challenges are not considered, such as interference management, channel estimation, and 3D placement of the UAV.	✓	×	×	×	×	×	×	×
	You et al. (2021)	Demonstrating the benefits of using UAV and RIS to enhance the system's efficiency.	Increasing the achievable rate of the system.	Interferences are not considered in the system's deployment.	×	×	×	✓	×	×	×	×
	Nnamani et al. (2021)	Maximizing the secrecy rate in a noisy communication channel in a UAV-RIS-based DC scenario.	Keeping a short distance with the BS while increasing the distance from the eavesdropper.	The UAV flight time negatively impacts the secrecy performance.	✓	×	×	✓	✓	×	×	×
RIS UAV cooperation	Ranjha and Kaddoum (2020)	Minimizing the decoding error rate and optimizing the UAV's positions and blocklength.	Achieving high reliability for short packets and increasing the RIS's performance gain.	The UAV and RIS energy consumption is not considered.	✓	×	×	×	×	×	✓	×
	Michailidis et al. (2021)	Improving end-to-end communication during the DC process.	A single RIS could significantly decrease the impact of phase errors on the system performance.	Multiple RISs are required to satisfy the requirement of a large-scale IoT environment.	✓	×	✓	✓	×	×	✓	×
	Cao et al. (2021a)	Improving the coverage of a UAV-assisted terrestrial system during the DC process using a RIS-based transmission strategy.	Increasing the system throughput and reducing the transmission power of devices.	The trajectories and movements of UAVs are not considered.	✓	✓	×	×	×	×	×	×
	Al-Hilo et al. (2022)	Maximizing the number of served IoT devices and improving the UAV's connectivity and energy efficiency.	The UAV energy consumption is significantly reduced.	It did not consider the freshness of collected data, which is inappropriate for large-scale scenarios.	✓	×	✓	✓	×	×	×	×
	Nguyen et al. (2022c)	Maintaining seamless connectivity of IoT devices while supplying them with energy.	Improving significantly the throughput and processing time performance of the system.	It is inappropriate for large-scale IoT environments since it operates with a single UAV.	✓	✓	✓	✓	×	×	×	✓
	Zhao et al. (2022)	Maximizing the energy efficiency of the system.	Decreasing the energy consumption of the whole system.	Did not consider the freshness of the collected information.	✓	✓	✓	✓	✓	×	×	×

coverage and capacity of the network is presented. Du et al. (2020) optimized the UAV trajectory and power allocation with massive MIMO hybrid beamforming to minimize the total transmit power under LoS channels. For this purpose, an approximate closed-form expression of the rate is derived. The authors of Feng et al. (2018b) assisted single antenna IoT devices using a multiantenna UAV for DC purposes. Thus, a virtual MIMO link is established to enable multiple access within a time-frequency resource block and adapt to the low transmit power of IoT devices while optimizing the transmission power of both IoT devices and the UAV. Table 13 provides a summary of these contributions.

Lessons learned

Despite the advantages of MIMO technology in terms of interference mitigation and coverage improvement, there are still certain drawbacks, such as the high energy consumption, inter-channel interference, and complex mechanism due brought on by multiple RF links (Canbilen, 2021; Ku, 2014). Moreover, when MIMO is involved in UAV-aided networks, it suffers from fading and signal degradation due to the high mobility of UAVs, which can reach long distances. Consequently, a number of solutions can be adopted to address the problems:

- Optimizing the transmission power of antennas and reducing the complexity of the overall system.
- Deploying efficient mechanisms to minimize inter-channel interferences.
- Establish a perimeter of movement for UAVs so that signal degradation will be reduced considerably.

4.8. NOMA for UAV-based data collection

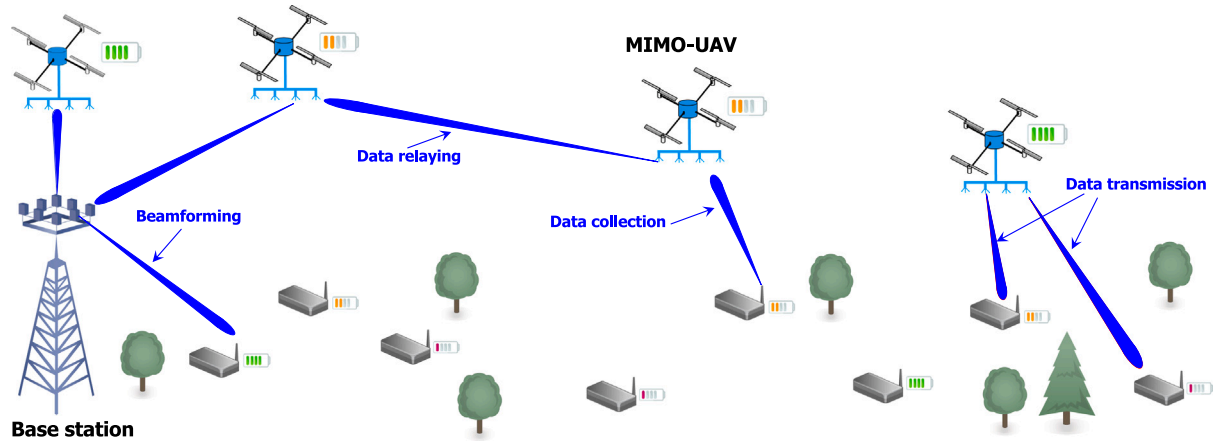
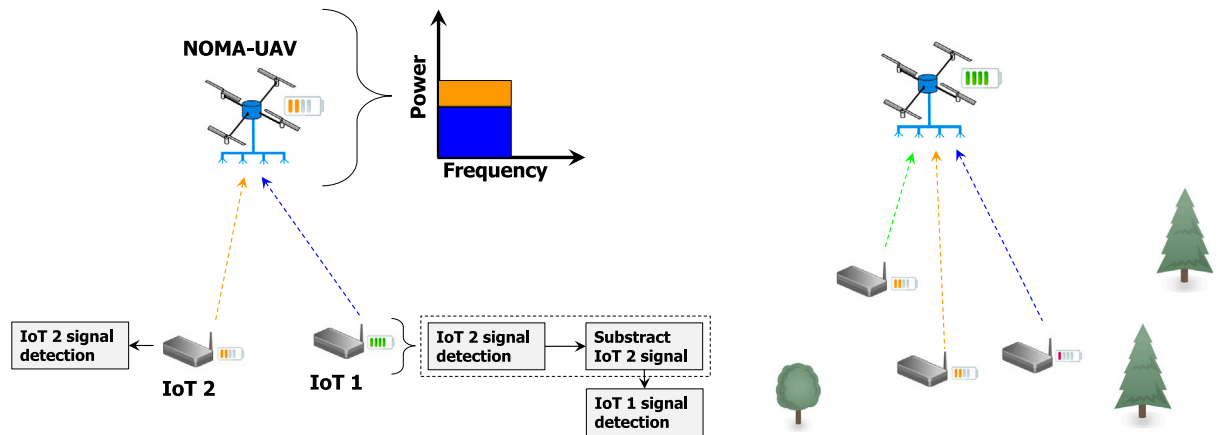
NOMA is a radio access technology to serve ultra-dense IoT networks with massive connections (Solanki et al., 2022). Indeed, as illustrated in Fig. 13, NOMA uses the same frequency resources to

establish connections between BSs and a large number of devices with different power transmission signals allowing an efficient signal compensation at the receivers (Jee et al., 2022). Therefore, adopting NOMA in the context of UAV-based DC is beneficial regarding latency and spectral/energy efficiency since UAVs have a restricted flying time that does not allow them to serve a large-scale IoT environment that requires a long data uploading time (Jee et al., 2021). For instance, in Wang et al. (2021c), a UAV-NOMA-based DC scenario is deployed where IoT devices are considered to be connected to the uplink NOMA network at each time-slot. The flight time, transmission power allocation, and UAV trajectory are jointly optimized. The authors of Mu et al. (2021) adopted uplink NOMA in energy-constrained UAV DC systems. Moreover, they optimized the UAV trajectory and the transmission power and scheduling of ground devices while considering the energy constraints of both UAVs and ground devices. Fu et al. (2022a) proposed a distributed gateway cluster algorithm to assemble gateways into multiple clusters based on the distances that separate them. Moreover, a UAV-BS is deployed over gateways to simultaneously collect their generated data based on NOMA. The purpose of this scheme is to maximize the energy efficiency of UAV-BS while considering the restricted flight duration, flight speed, and the transmitting power of gateways. The authors of Farajzadeh et al. (2019) adopted a UAV-NOMA model for assisting backscatter communication networks with the aim of maximizing the number of uplink bits that are successfully decoded. To this end, an optimization is jointly made for the UAV altitude, flight time, and spectrum efficiency. A minimization problem of energy consumption for UAV-NOMA-based DC system was investigated in Zhao et al. (2020). Indeed, since the system is deployed over a large-scale IoT environment and should ensure a certain level of QoS, a DC optimization scheme is applied, which is based on the generalized bender decomposition (GBD) and successive convex approximation techniques. Wang et al. (2022h) deployed a HAP and group of UAV data collectors above an IoT environment and with an eavesdropping UAV. The objective is to achieve a successful DC by improving the uplink secure rate by optimizing parameters, such as

Table 13

Summary of contributions of UAV-MIMO-based data collection.

		Objective	Advantage	Drawback	Cm	In	El	Mob	Se	Tsd	La	De
UAV-MIMO-based data collection	Hou et al. (2020)	Optimizing the beamforming transmission and energy efficiency of UAVs.	Providing low computational complexity and fast convergence.	High packet loss is distinguished and residual hardware impairments model is uncorrelated with input signal.	√	√	√	√	×	×	×	×
	Alkama et al. (2022)	Increasing the coverage and capacity of UAV-cellular communication networks.	the coverage probability provides good performances at different altitudes of UAVs.	The mobility of UAVs is not considered in this scheme.	√	√	×	×	×	×	×	×
	Du et al. (2020)	Minimizing the transmission power under LoS channels and thus reducing the energy consumption.	Maximizing the energy efficiency while ensuring the rate requirements for all devices.	The UAV position is not optimized and the system generates high complexity.	√	√	×	×	×	×	×	×
	Feng et al. (2018b)	Optimizing the transmit power of UAVs while being adapted to the limited transmission power of IoT devices.	Improving the efficiency of sensing data collection.	Only UAV circular trajectories are considered.	√	×	√	√	×	×	×	×

**Fig. 12.** UAV-MIMO-based DC.**Fig. 13.** UAV-NOMA-based DC.

the NOMA-enabled channel, IoT devices' transmit power, and UAVs' locations. We summarize this discussion in Table 14.

Lessons learned

In the literature, NOMA technology has demonstrated its reliability in providing higher spectral efficiency, supporting massive connectivity, and achieving better user fairness (Li et al., 2021a). Applying NOMA to UAV-based DC systems efficiently overcomes the spectrum scarcity by examining device heterogeneity in the power domain and enabling dense spectral sharing (Lu et al., 2022a). However, two main

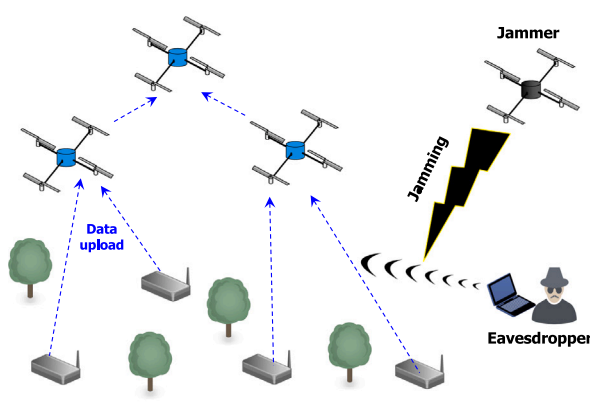
disadvantages are distinguished in NOMA: (i) increasing the computational complexity at the receivers due to successive interference canceler and (ii) higher sensitivity caused by inter-user interference. Consequently, some solutions are proposed to address these issues:

- Improving NOMA by increasing the number of transmission rounds.
- Enhancing the successive interference canceler.
- Using MIMO technology with NOMA to increase the spectral efficiency and connections again.

Table 14
Summary of contributions of UAV-NOMA-based data collection.

	Objective	Advantage	Drawback	Cm	In	El	Mob	Se	Tsd	La	De
UAV-NOMA-based data collection	Wang et al. (2021c)	Ensuring the freshness of collected based UAV-based DC system.	Generating successive interference cancellation with lower complexity under LoS condition.	The full potential of the UAV trajectory is not exploited with NOMA. Moreover, UAV energy consumption is not considered.	✓	✓	×	✓	×	×	✓
	Mu et al. (2021)	Maximizing the throughput of the UAV-based DC system using NOMA while considering the energy constraints of both UAV and IoT devices.	Maximizing the number of served IoT devices while increasing the data throughput.	Only UAV is considered, which is not suitable for large-scale IoT environment.	✓	✓	✓	✓	×	×	×
	Fu et al. (2022a)	Maximizing the UAV energy efficiency under the constraint of the restricted transmission power of gateways.	Decreasing significantly the UAV trajectory distance while achieving multiple access from gateways.	Limiting the number of users in a given cluster.	✓	✓	✓	✓	×	×	×
	Farajzadeh et al. (2019)	Maximizing the number of successfully decoded bits in the uplink while optimizing the UAV trajectory.	Improving the spectrum efficiency.	It assigns reflection coefficients statically.	✓	×	×	✓	×	×	×
	Zhao et al. (2020)	Minimizing the energy consumption of IoT devices while jointly optimizing IoT device resource allocation and UAV trajectory.	Providing better performance in terms of DC collection time.	It is not adapted for large-scale IoT environments.	✓	✓	✓	✓	×	×	×
	Wang et al. (2022h)	Maximizing the uplink secure rate while jointly considering UAVs' locations, IoT devices' power control, and the channel between UAVs and the HAP.	Providing a high uplink secure rate and fast convergence with less complexity.	Did not consider the energy consumption when deploying the proposed scheme.	✓	×	×	×	✓	×	×

Secure-based DC



Blockchain-based DC

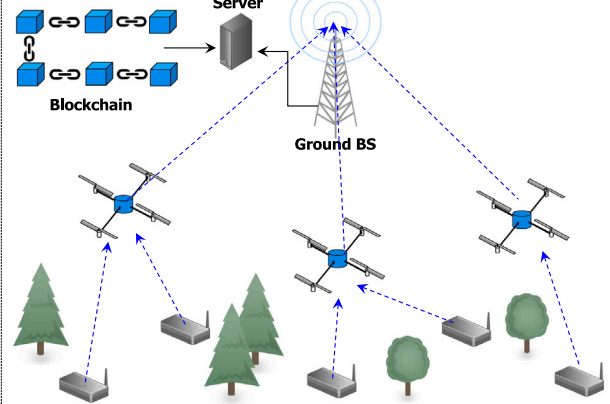


Fig. 14. Secure-based solutions for UAV-based DC.

4.9. Security in UAV-based data collection

UAV-based DC systems are frequently the target of cyber attacks that impact the UAV movement patterns, IoT devices, and communication strategies (Tsao et al., 2022). In the literature, we have distinguished that there are two categories of secure-based solutions (c.f., Fig. 14). Firstly, secure collection solutions, i.e., the different solutions employing security mechanisms to avoid certain attacks in the DC process. Secondly, blockchain-based solutions employ blockchain technology to secure the DC process. These two categories of solutions are presented in this section and summarized in Table 13.

4.9.1. Secure collection

An important number of security techniques are designed to protect UAV-assisted communications and face different cyber attacks (Xiao and Feroskhan, 2022). These techniques should have the capacity to ensure information confidentiality (Khan et al., 2021b), prevent outsiders from accessing the network (Masadeh et al., 2022), and authenticate

nodes (Zhou et al., 2022c). A set of secure-based contributions has been proposed in the literature to secure the UAV-based DC. For instance, Zhang et al. (2021) maximized the average secrecy rate in a dual-UAV communication system with multiple IoT devices. The authors optimized the propulsion energy consumption, IoT scheduling, and UAV trajectory. In Jiang et al. (2020a), the authors proposed a trust-based energy-efficient UAV-assisted DC method to increase the quality of DC and gather only trusted data, which are then used to optimize the UAV trajectory. The authors in Huang et al. (2020) proposed a joint mobile vehicle-UAV to compute the trustworthiness of collected data from sensors at the minimum flight cost. In Wang et al. (2020b), a UAV-based CH election scheme is proposed to enable secure sensor communications. Low energy consumption and security are ensured through the election of CHs with high residual energy, thus extending the lifetime of the system. Zhou et al. (2020b) studied a UAV-based confidential DC from scheduled sensors. The authors constructed an optimization problem by considering various measuring parameters, such as the secrecy and reliability outage probabilities.

4.9.2. Blockchain

Blockchain technology is considered as an efficient solution to guarantee secure communication and can be used in different applications (Li et al., 2021c). This technology consists of a series of interconnected blocks (*i.e.*, contain transactions) that are linked by the hash value of the block that came before it in the chain, and so on (Ali et al., 2021). UAVs can be outfitted with cryptographic methods using blockchain technology to provide secure communication (Mehta et al., 2020). For instance, Xu et al. (2020) deployed multiple UAVs to perform energy-efficient DC over terrestrial IoT devices. This scheme adopted blockchain technology to secure the DC, where honest UAVs are rewarded through virtual currency and an adaptive linear prediction technique to reduce overall energy consumption. In Li et al. (2021b), a blockchain-enhanced spatiotemporal UAV-based DC technique is proposed to secure the DC process. To establish a secure data transmission, the authors designed a UAV identity authentication mechanism based on a Merkle tree. Moreover, the redundancy in WSNs is reduced based on a sparsity-optimized and compressed sensing-based spatiotemporal data aggregation model. The authors in Islam and Shin (2019) used blockchain to secure the UAV-based DC over IoT networks where a bloom filter is used to validate the source's identity. Then, IoT devices transmit encrypted data to the UAV data collector, which is then transmitted by the UAV to the closest server, which prepares it for inclusion on the blockchain. To guarantee secure and confidential data monitoring, Xiao et al. (2021a) deployed a public key infrastructure (PKI)-based security protocol to protect management access to the monitoring data, authenticate participating UAVs and assign participating UAVs the monitoring duty. Then, the authors proposed to use a private blockchain with smart contracts to record the tasks' occurrences and activities. Similarly, the study in Lan et al. (2021) considered a UAV-assisted DC with blockchain-based security to secure IoT operations. With the IoT data collected by the UAVs, a blockchain with a proof-of-stake (PoS) consensus mechanism is specifically built. Then, the authors used a deep deterministic policy gradient-based approach (DDPG) to maximize IoT connection and UAV deployment for the highest blockchain throughput while taking the PoS method into account.

Lessons learned

UAV-based DC requires a certain level of security to achieve the DC process with no risk of jeopardizing the collected data. Across this survey, we discussed a set of secure-based schemes that tried to ensure the security requirement of the DC process. However, as in Zhang et al. (2021) and Huang et al. (2020), some secure-based schemes are not adapted to deal with multiple attackers, while others have only considered security parameters and neglected important communication parameters. Furthermore, the blockchain mechanism is regarded as resource-hungry for UAVs due to the amount of information that should be exchanged with centralized entities. To avoid such problems, we provide some recommendations that we think are flexible to be adopted in the context of UAV-based DC:

- Jointly considering the communication parameters with security-based principles.
- Adapting the secure-based schemes to general scenarios rather than focusing on specific techniques.
- Finding the appropriate methods to reduce energy consumption when applying the blockchain concept (see Table 15).

4.10. WPT in UAV-based data collection

WPT technology is attracting more and more interest in battery charging (Wang et al., 2022g). Indeed, this technology can be adapted to power UAVs and IoT devices and allow them to complete their missions without failures (*c.f.*, Fig. 15). A set of theoretical and experimental findings confirms the viability of using UAVs as energy transmitters

and wirelessly charging terrestrial sensors or IoT devices (Wang et al., 2022a; Jawad et al., 2022; Zhang et al., 2022). The contributions falling in this class can be classified into two categories: (i) powered UAV and (ii) powered IoT, which are discussed in this section and summarized in Table 16.

4.10.1. Powered UAV

UAV data collectors are well-known for their limited energy resources, which often cause their failures before the completion of their mission (Wang et al., 2022c). To avoid this issue, charging stations are deployed to assist UAVs during their mission and allow them to complete their mission (Arafat and Moh, 2022). A wireless-powered UAV is deployed in Fu et al. (2021) to gather data from IoT devices. During the DC process, the UAV trajectory is optimized using an RL strategy while considering its return to charging stations before energy depletion. Yan et al. (2021) studied the use of a UAV-enabled WPCN where the UAV is firstly powered by a BS and then deployed to make the WPCN process over IoT devices. This problem is described as a convex optimization problem with the objective of decreasing the UAV energy consumption, maximizing the energy transfer from the BS, and improving the data harvesting at the UAV and BS. The authors in Chen et al. (2019c) optimized both the trajectory of the UAV data collector and the transmission power of the charging station to extend the DC mission duration. Oubbati et al. (2022) deployed flying energy sources (FESs) to charge UAV data collectors and avoid disrupting their missions. A multi-agent deep RL (MADRL) is used to optimize the FESs' trajectories, maintain fairness when charging UAVs, and prevent collisions between FESs. In Liu et al. (2020c), a group of UAVs is deployed to perform the DC process while charging their batteries at multiple charging stations. The authors adopted an RL strategy to optimize the trajectory selection of each UAV and thus find the optimal solution for utility optimization. The authors of Du et al. (2019) deployed solar-powered HAPs and LAPs for DC purposes, where HAPs wirelessly charge LAPs. This architecture aims to maximize the sum-energy received by all LAPs by jointly optimizing the trajectories, energy consumption, and charging periods of LAPs.

It should be stressed that there are also two other methods to supply UAVs with energy: (i) the tethered method and (ii) the near-field charging method. The tethered approach is a classic method to provide a permanent energy source to UAVs (Lim et al., 2022). This is done by connecting the UAV to the charging station through a direct physical link (*i.e.*, cable) that allows the UAV to move in a pre-defined range. For instance, Saif et al. (2021) deployed a tethered UAV data collector to ensure its permanent energy supply throughout the mission. The objective is to optimize the coverage of the tethered UAV by optimizing its hovering angles, path loss, and coverage probability. As for the near-field charging approach, it is a bit different from traditional wireless charging through charging stations discussed above. Indeed, this approach can be based on other technologies, such as inductive charging or magnetic resonance charging, which could be integrated into charging stations that could be placed anywhere in the area of interest to allow UAVs to charge their batteries when needed by approaching a certain distance or landing above these stations (Wu et al., 2022a). For example, the authors of Yan et al. (2020) proposed a near-field WPT system where a UAV data collector could land or hover near the charging station to recharge its battery. The system has proven its efficiency by delivering an acceptable energy level at the highest distance compared to other methods.

4.10.2. Powered IoT

IoT devices are also considered as restricted energy devices that do not allow them to continuously perform their sensing mission, and transmission of sensed data towards data collectors (Yang et al., 2022a). As a solution, UAVs often play the role of charging stations to supply IoT devices with energy, and at the same time, UAVs collect data from them. This concept is known as a Simultaneous Wireless Information

Table 15
Summary of contributions of Security in UAV-based data collection.

		Objective	Advantage	Drawback	Cm	In	El	Mob	Se	Tsd	La	De
Secure collection	Jiang et al. (2020a)	Decreasing the UAV energy consumption by optimizing its trajectory while carrying out DC only from trusted sensors.	The system lifetime is significantly increased by 48.9% by optimizing the UAV trajectory.	The UAV failure is not studied, and no charging station deployment is studied.	×	×	✓	✓	✓	×	×	×
	Wang et al. (2020b)	Establishing an energy-efficient CH election and enabling secure communication among sensors.	Increasing the UAV movement improves the system's performance and security aspects.	Channel parameters are omitted, and the number of elected CHs is not carefully studied.	×	×	✓	×	✓	×	×	×
	Huang et al. (2020)	Enabling the UAV to collect more sensor-based baseline data to assess the trust of mobile vehicles (MVs) at the lowest flying cost.	Enhancing the trustworthy computation accuracy of sensors while increasing the data coverage ratio.	Focusing only on the trustworthy issue without taking into account other parameters and factors, such as throughput, the interaction between sensors and UAV, and packet losses.	×	×	✓	✓	✓	×	×	✓
	Zhou et al. (2020b)	Maximizing the minimum Average Secrecy Rate among all sensors.	Brought more gain in terms of secrecy rate with less complexity.	Did not work well with multiple UAVs due to the employed mechanism.	✓	✓	×	✓	✓	×	×	×
	Zhang et al. (2021)	Maximizing the secrecy rate by jointly optimizing the scheduling, trajectory, transmission power, and speed of the UAVs to avoid eavesdropping attacks.	Reducing the energy consumption of UAVs while maintaining the secrecy rate above a certain level.	Did not consider scenarios with multiple eavesdroppers.	✓	✓	✓	✓	✓	×	×	×
Blockchain	Xu et al. (2020)	Considering the security and energy efficiency of a UAV-based DC system using blockchain technology.	Ensuring high collected data accuracy while efficiently improving the security and energy efficiency of IoT devices.	The adopted mechanism is considered as resource-hungry for energy-restricted UAVs, and the rewarding mechanism under dynamic scenarios is missing.	×	×	×	×	✓	×	×	×
	Li et al. (2021b)	Ensuring the security and validity of the DC process while reducing data redundancy and establishing UAV identity authentication.	Increasing the network life cycle and data reconstruction accuracy.	Did not consider the UAV energy consumption, which could negatively impact the system operation.	×	×	✓	×	✓	×	×	×
	Lan et al. (2021)	Securing IoT operations through blockchain technology.	Optimizing the blockchain throughput while balancing the block propagation and transmission delay.	It did not consider the UAV energy consumption, and there is no collision avoidance mechanism.	✓	×	×	×	✓	×	✓	×
	Xiao et al. (2021a)	Ensuring the confidentiality and security of the UAV-based monitoring data.	Increasing the authenticity of the UAV identity, the UAV cooperation reliability, and the collected information confidentiality.	The key establishment step considerably increases the delay, among other processes.	×	×	×	×	✓	×	×	×
	Islam and Shin (2019)	Recording collected data in the nearest servers with the assistance of blockchain.	Extending the connectivity and reducing the energy consumption of IoT devices.	The system is more access friendly due to the lack of private blockchain.	×	×	✓	×	✓	×	✓	×

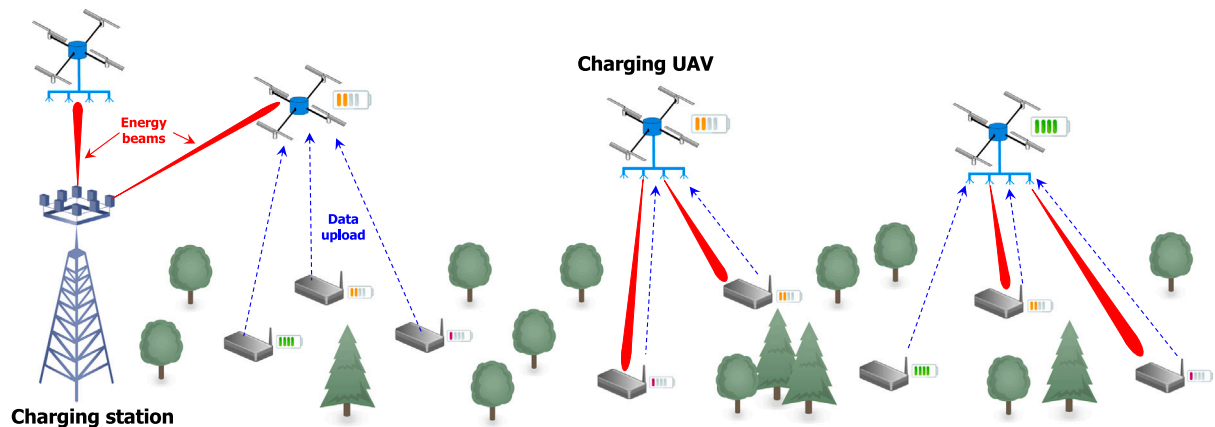


Fig. 15. Wireless powered UAV and IoT for DC purposes.

and Power Transfer (SWIPT) (Liu et al., 2022). For instance, in Feng et al. (2020), a UAV-enabled SWIPT scheme for the IoT environment is proposed where the UAV trajectory, the IoT device charging time, and the resource scheduling are jointly optimized. Moreover, a UAV path planning problem is studied to provide connectivity for IoT devices. The aim of this scheme is to maximize the whole system's energy efficiency.

Uplink data collection and downlink wireless energy transfer (WET) are used simultaneously in Li et al. (2019a). These two tasks are optimized through a DQN method to minimize packet losses while enhancing IoT device charging, data gathering, and preventing data queue overflow and battery drainage of IoT devices. Another strategy is also adopted when powering IoT devices where the UAV first charge IoT devices,

Table 16
Summary of contributions of WPT in UAV-based data collection.

		Objective	Advantage	Drawback	Cm	In	El	Mob	Se	Tsd	La	De
Powered UAV	Fu et al. (2021)	Improving DC from sensors while increasing the energy efficiency of the system.	Achieving better performance in terms of throughput, UAV energy consumption, and DC delay.	A single UAV data collector is deployed, which cannot meet the requirements of a high-scale IoT environment.	√	×	√	√	×	×	×	×
	Yan et al. (2021)	Maximizing the amount of collected data offloaded at the BS while considering the fixed total time allocated for this mission.	Increasing the amount of data at the BS with optimal use of the allocation time.	Did not consider the different channel conditions and their impact on the DC process.	√	×	√	×	×	×	×	×
	Liu et al. (2020c)	Efficiently charging a group of UAV data collectors from randomly deployed charging stations.	Increasing the energy harvesting from charging stations while ensuring DC efficiency.	Taking too much time to converge while neglecting the utility functions of each entity.	×	×	√	√	×	×	×	×
	Oubbati et al. (2022)	Achieving appropriate fairness in UAV charging.	Obtaining a significant energy transfer gain by optimizing the FESs' trajectories and energy consumption.	Interferences and other physical phenomena are not considered.	√	×	√	√	×	×	×	√
	Chen et al. (2019c)	Maximizing the energy transmission efficiency by optimizing the UAV trajectory and the charging station's power transmission.	Improving the UAV endurance, and thus extending their maximum mission duration.	Did not explicitly consider the UAV's energy consumption. Moreover, there is only a single charging station, which cannot satisfy the energy requirements of multiple UAVs.	√	×	√	√	×	×	×	×
	Du et al. (2019)	Minimizing the quantity of laser energy for charging HAP while optimizing the LAP trajectory and charging duration for each LAP.	The optimization method quickly converges to optimal performance.	Solar energy is insufficient to power HAPs, so the scheme is unrealistic.	×	×	√	√	×	×	×	×
	Saif et al. (2021)	Increasing the coverage area of the UAV in the context of disaster recovery.	Providing good performances in terms of coverage probability.	The UAV movement is restricted by the tether and the area of interest is limited.	×	×	×	×	×	×	×	×
Yan et al. (2020)	Providing an acceptable WPT efficiency under the constraint of UAV mobility.	Achieving high charging efficiency.	The UAV landing schedule is not considered.	×	×	×	×	×	×	×	×	
Powered IoT	Li et al. (2019a)	Minimizing the packet losses of IoT/sensor devices while determining which device should be charged and interrogated most efficiently.	Decreasing the packet loss by at least 69.2% by jointly optimizing the UAV trajectory and charging.	It did not consider the UAV energy consumption and interferences that may occur during the DC process.	√	×	×	√	×	√	×	×
	Feng et al. (2020)	Dynamically planning UAV paths to increase the energy efficiency of the system.	Achieving timely DC service for IoT devices while avoiding running out of UAV's battery.	It did not satisfy a large-scale IoT environment and did not consider any collision avoidance mechanisms.	×	×	√	√	×	×	×	×
	Liu et al. (2019c)	Jointly optimizing resource allocation for both power and information transmission of UAVs and IoT devices and UAV trajectory.	Harvesting more energy to make more data transmission towards UAVs.	Did not consider the association problems of UAVs with ground devices.	√	√	√	×	×	×	×	×
	Chen et al. (2019a)	Decreasing the transmission completion time of collecting a given amount of data per IoT device.	Significantly reducing the DC with a large number of ground devices.	The UAV is deployed at a fixed position and do not move around IoT devices.	√	×	√	×	×	×	×	×
	Wang et al. (2019d)	Increasing the energy harvesting efficiency and avoiding channel fading.	An efficient balance between the energy harvesting and transmission phases based on the optimal UAV trajectory.	Satisfying a reduced number of IoT devices.	√	×	√	√	×	×	×	×

and then the harvested energy is used by IoT devices to transmit their generated data towards UAVs. This concept is known as a Wireless Powered Communication Network (WPCN) (Hashir et al., 2021). For instance, Liu et al. (2019c) applied a UAV-enabled WPCN over an IoT environment where the dynamic game theory is adopted to optimize the resource allocation in terms of energy according to the density of IoT devices. The authors in Wang et al. (2019d) deployed a UAV to charge IoT devices and perform DC. A Successive Convex Approximation (SCA) is adopted to jointly optimize the IoT devices' scheduling (i.e., time allocation for data transmission) and UAV trajectory, with the aim to maximize the minimum throughput of IoT devices and reduce the UAV energy consumption. To minimize the completion time of a UAV-enabled WPCN mission and collect a given amount of bits from

each IoT device, the authors of Chen et al. (2019a) adopted both a bisection search and subgradient algorithms to solve this optimization problem.

Lessons learned

Charging UAVs or ground devices during DC missions requires the proper use of WPT technology. In this survey, we distinguished some issues when charging UAVs, such as the placement of charging stations, their density, and interferences. Furthermore, collisions, large-scale scenarios, and UAV trajectories are not adequately considered when charging IoT devices. For instance, in Feng et al. (2020), no collision avoidance mechanisms are considered, and UAV charging stations cannot serve large-scale IoT devices. However, all these issues

Table 17

Simulation tools for evaluating UAV-based data collection schemes. (N/A) shows that no comparable sources were considered by the authors. (×) shows that there are no simulation tools stated or utilized.

	References	Calculated metrics	Compared references	Simulation tool(s)
AI	Li et al. (2020c)	QCD	N/A	Python 3.5, TensorFlow with Keras
	Xiong et al. (2020)	EC, EE, TDC, Tp	Baseline methods	×
	Oubbati et al. (2021)	EC, AoI	Baseline and RL methods	Python 3.5, TensorFlow with Keras
	Nguyen et al. (2022a)	ADR, EC, Tp	RL methods	Python with Adam optimizer
	Wang et al. (2021a)	TDC, EC, QCD	Dorigo et al. (1996), Baseline method	Python 3.6 and PyTorch 1.4
	Li et al. (2019b)	TDC	Baseline method	Matlab®
	Chen et al. (2019b)	–	RL methods	Python 3.6, TensorFlow
UAV Control	Yang et al. (2023)	–	AI methods	Scikit-Learn in Python 3.6
	Gong et al. (2018)	TDC	Customized methods	Matlab®
	Zhou et al. (2018a)	EC	Benchmark methods	×
	Lu et al. (2022b)	EE, EC, ADR	Baseline methods	Matlab®
	Benmad et al. (2021)	TDC	Benchmark methods	×
	Zhan and Zeng (2019a)	–	Customized and benchmark methods	Matlab®
	Shen et al. (2021b)	–	Benchmark methods	Matlab®
	Liu et al. (2019b)	EC, QCD	Customized methods	VC++
	Samir et al. (2019)	EC, ADR	Baseline methods	×
	Liu et al. (2020d)	ADR, QCD, EC, Tp	Baseline methods	C++
	Zhan and Zeng (2019b)	TDC	Baseline methods	Matlab®
	Goudarzi et al. (2019)	ADR, EC	Ho et al. (2015), Baseline methods	OMNeT++, Matlab®
	Gao et al. (2023a)	–	Mellinger and Kumar (2011), Richter et al. (2016)	TestBed
	Qayyum et al. (2021)	EC, TDC	Yang et al. (2018a)	OMNeT++
	Tao et al. (2019)	EC, TDC	Customized methods	×
	Wang et al. (2020d)	TDC	Customized methods	Matlab®
	Pěnička et al. (2019)	–	Customized methods	CPLEX 12.6.1
Data Transfer	Asadi et al. (2020)	–	N/A	TestBed
	Tazibt et al. (2018)	TDC, QCD	Pereira et al. (2004)	×
	Baek et al. (2019)	EC	Shmoys et al. (1985), He et al. (2011)	×
	Say et al. (2016)	EC, ADR, Tp	Heinzelman et al. (2000), Baseline method	Matlab®
	Okcu and Soyuturk (2014)	EC, EE	Baseline methods	OMNeT++
	Ho et al. (2015)	EC, ADR, TDC, BeR	Heinzelman et al. (2000)	Monte-Carlo
	Ebrahimi et al. (2018)	EC, BeR, EE	Baseline methods	CPLEX
	Rozaliya et al. (2022)	QCD, Tp	Baseline method	GUROBI
	Liu et al. (2018c)	AoI, BeR, OP	Baseline methods	Matlab®
	Abdulla et al. (2014)	EC, ADR, Tp, BeR	Baseline method	Customized simulator
	Jawhar et al. (2015)	Tp	N/A	×
	Ahmed et al. (2020)	EC	Baseline methods	Matlab® with CVX
	Araghizadeh et al. (2016)	ADR, EC	Ho and Shimamoto (2011)	C#
	Mori (2016)	EC	Erceg et al. (1999)	Digi International's XBee
	Sotheara et al. (2014)	ADR, QCD, TDC, Tp, BeR, EE	Baseline method	Matlab®
	Zanjie et al. (2014)	ADR, Tp, BeR	Baseline method	×
	Vlăduță et al. (2016)	QCD, EE	N/A	NS-2
Time-sensitive data	Ruan et al. (2020)	Tp, ADR	Wu et al. (2018a)	×
	Hu et al. (2020)	AoI, QCD, TCD	Yang et al. (2018b), Boshkovska et al. (2015, 2017), Valenta and Durgin (2014)	×
	Wang et al. (2022f)	AoI, ADR	Jella and Sabat (2018)	Wei et al. (2020)
	Jia et al. (2019)	TCD, AoI	Baseline method	×
	Tong et al. (2019)	TCD, AoI	Baseline methods	×

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could be avoided if the researchers had taken the required measures. Therefore, as an effort on our part, we provide some ideas to prevent these issues as follows:

- Designing the most appropriate channel models to deploy the WPT concept while avoiding interferences and collisions.
- Optimizing the density and placement of charging stations. Moreover, mobile charging stations could also be an excellent option to avoid deploying many charging stations.
- Optimizing the trajectories of UAVs and their charging scheduling when deployed as charging stations.

5. Common myths and critical questions

This study illustrates that the UAV-based DC concept is founded on a well-settled theory. However, the state-of-the-art of this research domain has been subject to several prevalent myths and unanswered questions. A deep discussion is needed here to face the reality of things

and bring the UAV-based DC concept from theory to practice on a solid basis. Therefore, in this section, we analyze the seven (07) major false beliefs and provide arguments to deny them.

Myth 1: UAVs are not human operated

There is a widespread misconception that UAVs are not human-operated. Indeed, in most UAV-based DC solutions, UAVs are supposed to be controlled through a centralized entity (agent) or a base station via a backhaul connection during the whole mission. Moreover, in some instances, UAVs operate in a distributed manner with any centralized control where each UAV is controlled by its respective agent embedded in it and executes, in parallel, many algorithms, such as path-planning methods, resource allocation management, and communication protocols. However, the question that arises here is “*Could the human operator be useful in UAV control?*”.

There are very few UAV-based DC systems that involve humans in some way during their functioning. Human involvement is necessary

Table 17 (continued).

	References	Calculated metrics	Compared references	Simulation tool(s)
	Wu et al. (2021c) Abd-Elmagid et al. (2019) Mao et al. (2020) Liu et al. (2020e) Wu et al. (2021a) Ghdiri et al. (2021)	AoI, Tp AoI AoI AoI TCD, AoI EC, QCD, AoI	Wu et al. (2020) and baseline methods Baseline methods Baseline methods Lyu et al. (2016) and baseline methods Whitley (1994) and baseline methods Alfattani et al. (2019) and baseline methods	× Python with TensorFlow × × PyTorch ×
	Wang et al. (2018) Popescu et al. (2020) Singh and Sharma (2022)	TCD Crop information TCD and Crop information	N/A N/A Spachos and Gregori (2019), Raeva et al. (2019)	TestBed NMEA (NMEA Generator, 2023) TestBed
	Tsoursos et al. (2019) Zhou et al. (2021) Andreadis et al. (2022)	Crop information EC, ADR OP	N/A Shah et al. (2012) Customized methods	TestBed Matlab® Matlab®
Data collection applications	Poudel and Moh (2021) Duong et al. (2019) Erdelj et al. (2017) Wang et al. (2020a)	QCD, EC, TDC TDC EE QCD, EC	Duan and Li (2015) N/A N/A Lalouani et al. (2018), Lee et al. (2018), Python Liu et al. (2018a) Liu et al. (2012)	Matlab® TestBed × ×
Emergencies	Demiane et al. (2020)	Disaster information, EE		×
	Shen et al. (2021a) Zhang et al. (2017) Huang et al. (2021a) Lee et al. (2021) Tan et al. (2019)	QDC QCD ADR, QCD Revenue of the system, Tp EC, QCD, Tp	Baseline methods Baseline methods Baseline methods Baseline methods Baseline methods	C++, Python Python 3.5 with TensorFlow Matlab® PyTorch Matlab® with CVX
	Wang et al. (2019b) Kim and Qiao (2020) Qin et al. (2019) Wang et al. (2020d) Zhan et al. (2017) Zhu et al. (2018) Nazib and Moh (2021)	EC, EE, OP EC, EE, QCD QCD TDC, Tp EC, EE, OP TDC, QCD, Tp, BeR TDC, EC, EE	N/A Implemented Baseline methods Baseline methods Baseline methods Implemented baseline methods Customized methods Younis and Fahmy (2004), Heinzelman et al. (2002)	Matlab® Matlab® with CVX and Gurobi × Matlab® × × ×
Energy Efficiency	Ghdiri et al. (2020) Dong et al. (2019) Wu et al. (2019) Jia et al. (2021)	EC, ADR EC, Tp ADR, EC, TDC, EE, Tp ADR	Alfattani et al. (2019), Nielsen (2016) Customized schemes Wu et al. (2018b), Liu et al. (2013) Benchmark and customized methods	× × OMNeT++ Matlab®
	Mahmoud et al. (2021) You et al. (2021) Nnamani et al. (2021)	ADR, OP ADR ADR and security information	Baseline methods N/A Cui et al. (2019) and a proposed method	Matlab® Matlab® ×
UAV-RIS based DC	Ranjha and Kaddoum (2020) Michailidis et al. (2021) Cao et al. (2021a) Al-Hilo et al. (2022) Nguyen et al. (2022c) Zhao et al. (2022)	ADR EC, TDC ADR, Tp EC, ADR, EE ADR EC	Baseline methods Baseline methods Baseline methods Baseline methods Customized methods Customized methods	× Matlab® with CVX × Python with TensorFlow Python with TensorFlow ×
UAV-MIMO based DC	Hou et al. (2020) Alkama et al. (2022) Du et al. (2020) Feng et al. (2018b)	EC ADR, OP ADR ADR	Customized schemes Customized schemes Chen et al. (2017b) Customized methods	Matlab® Matlab® Matlab® Matlab®
UAV-NOMA based DC	Wang et al. (2021c) Mu et al. (2021) Fu et al. (2022a)	ADR, TDC, QCD ADR EE, TDC	Baseline methods Customized methods Hong and Kwong (2008), Shen et al. (2016)	Matlab® Matlab® Matlab®
	Farajzadeh et al. (2019) Zhao et al. (2020) Wang et al. (2022h)	TH, BER EC Tp	Customized methods Customized methods Benchmark methods, Pischella et al. (2020)	× Matlab® Matlab® with CVX

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in the sense that it can be helpful in case of a malfunction where the human pilot can take over this situation and avoid any incidents that expose people or materials to danger. Consequently, in any realistic UAV deployment, a human operator is always present in the area to intervene when any issues occur.

Myth 2: Deployment environments do not impact the performance of UAV-based DC systems

This misunderstanding is founded on two false hypotheses. First, it is frequently supposed that the weather is clear in the deployment

environments, which is not, however, correct given that there is at least a non-negligible wind speed. Second, it is also assumed that IoT devices or sensors are deployed on flat terrain on the ground, which is also incorrect since these devices are generally deployed in places with non-flat terrains, such as forests, disaster areas, or even mountains. Therefore, a critical question is: “Can deployment environments impact negatively the performances of UAV-based DC systems?”.

The weather greatly impacts the functioning of any UAV-based systems in the sense that it could increase or decrease the energy consumption of UAVs when the wind is favorable or unfavorable, respectively. Moreover, rain can also be a factor that can increase

Table 17 (continued).

	References	Calculated metrics	Compared references	Simulation tool(s)
Secure DC	Jiang et al. (2020a)	EC and security metrics	Customized methods	Python
	Wang et al. (2020b)	QCD and security metrics	Customized schemes	NS-2
	Huang et al. (2020)	ADR	A customized scheme	Matlab® with T-Drive dataset (Yuan et al., 2011)
	Zhou et al. (2020b)	ADR, OP, and security metrics	Customized schemes	Matlab®
Secure Blockchain	Zhang et al. (2021)	ADR, Tp, and security metrics	Customized schemes	×
	Xu et al. (2020)	EC, EE and security metrics	Raza et al. (2015)	Matlab®
	Li et al. (2021b)	QCD, EC	Customized schemes	TestBed
	Lan et al. (2021)	ADR	Customized methods	Python with TensorFlow
WPT	Xiao et al. (2021a)	TDC and security metrics	Customized schemes	A customized simulation tool
	Islam and Shin (2019)	EC, QCD, ADR	Ullah et al. (2019), Wang et al. (2017) and customized schemes	Matlab®
Powered UAV	Fu et al. (2021)	QCD, EC, TDC, EE, Tp	A customized scheme	×
	Yan et al. (2021)	ADR, QCD	Customized schemes	Matlab®
	Liu et al. (2020c)	EC, QCD, EE	Liu et al. (2019a)	PyTorch
	Oubbati et al. (2022)	EC, QCD	Baseline methods	Python with TensorFlow
Powered IoT	Chen et al. (2019c)	TDC, EC, OP	Customized schemes	Matlab®
	Du et al. (2019)	EC	Benchmark methods	Matlab®
	Saif et al. (2021)	OP	Customized methods	×
	Yan et al. (2020)	–	Customized methods	TestBed
	Li et al. (2019a)	QCD, ADR, OP	Baseline methods	Python with TensorFlow
	Feng et al. (2020)	ADR, QCD, EC, EE	N/A	Matlab®
	Liu et al. (2019c)	EC	A customized method	Matlab®
	Chen et al. (2019a)	TDC, EE	Customized methods	×
	Wang et al. (2019d)	ADR, EC	Baseline methods	Matlab®

UAV energy consumption since it can weigh down UAVs that could respect a given altitude. Consequently, the weather is a factor that should absolutely be taken into account. As for IoT/sensor deployment, neglecting the deployment terrains of these devices could significantly distort the modeling of communication channels between them and UAVs. Furthermore, induced errors in the UAV-IoT/sensor links could negatively impact the DC process and, thus, decision-making carried out in the system's backend. Therefore, the terrain should be carefully studied beforehand to generate realistic and correct results.

Myth 3: The data generation of IoT devices or sensors is low

Most authors suppose that IoT devices or sensors generate a reduced amount of data to be collected. Furthermore, the frequency of data generation is also assumed to be low. These two hypotheses are not realistically correct since IoT/sensor devices are generally deployed to permanently monitor the deployment area. The question that has to be asked here is: “Are UAVs perfectly suited for collecting a huge amount of generated data?”.

Indeed, in certain cases, IoT/sensor devices generate data at a rate similar to a mobile video stream consistently, *i.e.*, 24 h a day and 365 days a year. This data generation frequency causes, what we call, a bandwidth-hungry situation that significantly impacts network communications. Consequently, future UAV-based DC solutions should analyze the frequency of data generation.

Myth 4: UAVs finish their missions without failures

The performance evaluation of most UAV-based DC solutions demonstrates the success of the UAV mission in performing the DC process without any failures. In reality, deployed UAVs can face many problems and challenges which cause frequent UAV failures before obtaining suitable trajectories. For instance, UAVs could be disconnected from the centralized controller at any time due to bugs or poor radio conditions causing UAV failures. Furthermore, multiple tests are required to evaluate the performances of UAV-based systems, which causes energy depletion due to excessive energy consumption. Also, collisions with obstacles due to unpredictable events could cause UAV failures. The question here is: “How to avoid UAV failures in critical missions?”.

UAV failures frequently occur during experiments. Therefore, it is very important to analyze the different causes of failures to avoid them during real missions where no failures are tolerated. Moreover, the number of failures during the experiment could be considered as a metric to show the effectiveness of the architecture as well as the proposed UAV-based DC solutions.

Myth 5: AI provides always good performances in UAV-based DC

AI is increasingly present as an optimization method in UAV-based DC systems. Converging AI-based solutions often takes a lot of time, depending on the environment's dynamic or the correctness of the used datasets. Furthermore, since the learning phase frequently fails to consider all possible situations of the environment, the evaluated system could generate non-optimal decisions, thus decreasing its performance. The critical question here is: “Can UAV-based DC systems benefit from AI?”.

The objective of AI-based solutions is always to make near-optimal actions or decisions. Since we deal with time-varying communication channels in UAV-based DC systems and highly dynamic network topology, the learning phase should continue even after the realistic deployment of the systems to learn unknown situations not faced during the initial learning phase. Consequently, AI is an exciting research direction for which large-scale studies can derive the progress of this domain.

Myth 6: Embedding RIS in UAVs is beneficial for DC

Involving RIS technology in UAV-based DC systems is not a new concept due to its several benefits. In particular, the systems using MIMO technology and mmWave communications took the plunge towards adopting RIS technology. However, a RIS is an equipment that requires to be supplied with energy, especially when it is mounted on UAVs. Moreover, its weight is not negligible, also impacting the energy consumption of UAVs when it is mounted on them. The important question that should be asked here is: “is it beneficial for UAVs to be mounted with RISs?”.

Even if RIS technology has demonstrated its efficiency in interference mitigation and improved spatial multiplexing, the energy consumption issue is still a challenge. Therefore, a deep investigation is required to analyze the energy consumption of RISs in terms of

reconfigurability and control channels before mounting such equipment on UAVs.

Myth 7: WPT technology is the appropriate energy supply for UAVs

WPT technology has become an attractive solution to avoid the energy constraints of UAVs. Indeed, it is a technology that can charge the batteries of UAVs wirelessly without any human intervention. However, a set of drawbacks are easily distinguished, such as the low energy transfer efficiency and difficulty in maintaining constant load impedance. These drawbacks are due to the mobility of UAVs to be charged and the blockage caused by the different obstacles. The critical question here is: “Is WPT technology the appropriate solution to power UAVs?”.

WPT technology is based on a straightforward and safe concept and can quickly adapt to the deployment environment. Nevertheless, the drawbacks of this technology do not allow it to be deployed realistically before efficiently overcoming them. Therefore, there are alternative energy charging solutions that demonstrate their efficiencies, such as the tethered energy supply, solar energy, and optimized battery replacement processes.

6. Simulation tools and evaluations

In reality, UAV-based data collection is multidisciplinary, requiring adaptable methods to assess their effectiveness and the simulation tools that go with it. These simulation tools should also be flexible in various areas, including network design, communication systems, optimization theory, and relevant topics like game theory, AI and machine learning, and stochastic geometry. In this section, we focus on the simulation tools used to assess the performance of the proposed solutions outlined in Section 4. This section is divided into two subsections dealing with a summary of evaluation tools and statistical analysis.

6.1. Statistical analysis

To evaluate the performance of the UAV-based schemes. Statistical analysis is carried out on the most used simulation tools as depicted in Fig. 16. The statistics are generated based on the UAV-based DC schemes discussed in this survey where the different percentages are calculated as $(\frac{N_i}{R})$, where N_i is the number of times where the evaluation tool “ i ” is used in the considered UAV-based DC schemes and R the total number of UAV-based DC schemes discussed in this survey. This figure shows that most researchers adopted mainly simulation tools and programming languages, such as Matlab®, Python, OMNeT++, and homemade tools to evaluate their proposed UAV-based DC solutions. It should be stressed that there are several challenges when performing the simulations, such as UAV deployment security, environmental factors, UAV communication issues, infrastructure costs, and the complexity of data extraction from the network. To perform realistic trials, it is necessary to address each of these limits and challenges. For this purpose, another way is to develop alternative and unified UAV simulation tools that would help researchers accurately analyze various UAV-based schemes based on their hardware features.

6.2. A summary of evaluation tools

Table 17 summarizes the simulation tools used for the UAV-based data collecting schemes discussed in our survey. In addition, we provide the key evaluation metrics used in each case, which are already described in Section 2.3.

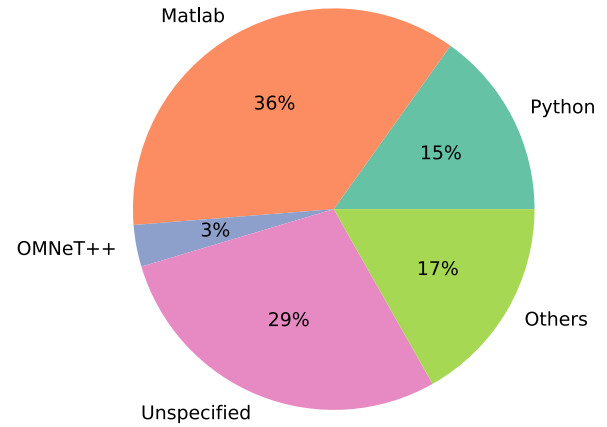


Fig. 16. Statistics about simulators and evaluation tools.

7. Open problems and future research perspectives

As was already discussed, our survey highlights several significant research issues to consider for UAV-based DC schemes' future design. The research on UAV-based DC is still in its infancy, and the distinguished problems have to be thoroughly explored to allow UAV data collectors to operate flexibly and efficiently using their physical resources. In this section, we analyze several less-investigated research directions that require further analysis and attention from the scientific community, such as the integration of AI, Blockchain technology, spectrum management, Beamforming technology, RIS, and interferences in massive UAV-assisted networks. Moreover, we present some potential solutions and recommend interesting references to follow, all included in Table 18.

7.1. AI for UAV-based DC

There are several issues in UAV-based DC schemes which may not be solvable using conventional model-driven techniques, mainly when the UAV-based DC process is carried out in unknown environments. For instance, the user association process to make DC based on UAVs demonstrates complicated properties, such as randomness of data generation, lack of energy and drone coverage, scalability, the freshness of generated data, and distributed decision-making. Furthermore, the trajectory and placement of UAVs face various constraints, such as the minimum/maximum flying altitude and speed, moving or static obstacles, restricted residual energy, mission duration, and resource restrictions in frequency. Also, the different areas of UAV-based DC applications suffer from several challenges, such as the density of aerial/ground sensors to be deployed in agricultural fields, UAV failure in emergency applications, and wireless environmental conditions in a smart city. Finally, the UAV-RIS-based DC concept introduces several challenging issues involving the combined optimization of the signal transformations done by the RISs and the UAV trajectories. As a solution, AI is considered a potent strategy to overcome the above-mentioned challenges and optimally learn the environments where the DC process is performed.

7.2. UAV-RIS-assisted DC

When a RIS is embedded in UAV data collectors and assists them in conducting DC, multiple challenges arise due to different issues. First, a RIS device requires a continuous energy supply, which is difficult to perform using the restricted energy of UAVs. Second, it is very complex to optimize at the same time the UAV trajectory and the phase shift design of RISs due to their different types of state spaces. Third, channel conditions are not always considered by RISs, which

Table 18

Open research directions for UAV-based DC.

	Problems	Proposed solutions	Recommended references
AI for UAV-based DC	- The user association process is not optimally scheduled due to different issues. - The trajectory and placement of UAVs are not optimized based on the dynamics of environments. - UAV-based DC applications focused mainly on data-driven optimizations while neglecting environmental conditions and the restricted energy of nodes. - Difficulties in implementing the concept of UAV-RIS-based DC due to the trajectories of UAVs and the signal transformations performed by the RISs.	Machine and deep learning techniques can be used to learn the dynamics of the environments, the random specificity of specific user association protocols, and the complexity of directed signals.	Lahmeri et al. (2021), Rezwan and Choi (2022), Parvaresh et al. (2022), Rovira-Sugranes et al. (2022), McEnroe et al. (2022), Bithas et al. (2019), Taha and Shoufan (2019)
UAV-RIS-assisted DC	- Lack of energy from RIS when integrated into UAVs. - Difficulty optimizing the phase shift design of RISs and the UAV trajectories in parallel. - Neglect of wireless channel conditions by RISs causes abnormalities in their functioning. - Most UAV-RIS-based DC schemes are not adapted to large-scale IoT environments.	To address these issues, we propose to leverage WPT technology to supply RISs with energy, use AI-empowered methods to optimize both the RISs' phase shift design and UAV trajectories, consider the channel conditions by using Beamforming technology and virtual LoS path, and accurately estimate the RISs' density to optimally serve large-scale IoT environments.	Liu et al. (2021a), Ye et al. (2022), Mohsan et al. (2022), Park et al. (2022), Pogaku et al. (2022)
Beamforming for UAV-based DC	- Different issues in the terrestrial environments could disturb mm-Wave UAV communications, such as the reflections, static/moving obstacles, and orientation of devices. - The performance metrics of mm-Wave UAV communications are not unified and defined beforehand. - The resource allocation in the cross-layer is not carefully studied. - Interference issues of mm-Wave communications when several communications are established in parallel.	Beamforming technology and accurate beam alignment are efficient for mm-Wave UAV communications. Furthermore, various optimization and AI methods could be adopted to quickly respond to the environment's dynamics and select optimal beams based on the environment states.	Xiao et al. (2021b, 2020), Zhang et al. (2019b), Li et al. (2018), Sharma et al. (2020)
Interference in Massive UAV-assisted networks	- Future terrestrial networks require more and more densities of UAVs, causing a high level of interferences, and thus negatively impacting the network and transmission's performances. - A shared wireless channel by UAVs is constantly exposed to unavoidable and complex interferences. - The exchange of information between UAVs and terrestrial macro cells could cause interferences. - The coverage of a UAV could overlap with another, causing interferences.	Due to the dynamics of interferences within massive UAV-assisted networks, learning techniques could be the appropriate solution to significantly decrease the level of interferences by reducing information exchange between UAVs, increasing the autonomy degree of nodes, and enhancing the self-organization capacity of the networks.	Challita et al. (2019), Pang et al. (2022), Li et al. (2022d), Azari et al. (2022), Fouda et al. (2019), Rahmati et al. (2021), Pi and Zhou (2021), Tang et al. (2021)
Resource and spectrum scheduling for UAV-based DC	- Spectrum scheduling overlooked crucial parameters to increase its efficiency. - Security and interference issues are not considered when establishing backhauling through UAVs. - High relay energy consumption and delays are distinguished when realizing offloading through UAVs. - The placement of UAVs and the UAV density to be deployed are not appropriately estimated and optimized due to the neglect of various restrictions, such as environmental obstacles, mobility of ground devices, and energy consumption of UAVs.	Several solutions could be adopted to address the mentioned issues, including optimization and AI methods, while considering, at each time, the appropriate metrics. Therefore, these solutions are worthwhile to further investigation.	Feng et al. (2018a), Wang et al. (2022b,e), Jasim et al. (2021), Kakar and Marojevic (2017), Qamar et al. (2020), Masroor et al. (2021)

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considerably disturbs their functioning. Finally, it has been shown that optimizing the deployment of RISs in large-scale IoT environments is challenging due to the restricted number of elements composing the RISs. Several solutions are possible to address these issues, including the use of WPT technology to power RISs, AI-empowered techniques to optimize both UAV trajectories and RISs' phase shift design, Beamforming technology and a virtual LoS path to enhance the channel conditions, and optimization of RISs' density estimation to optimally serve IoT environments.

7.3. Beamforming for UAV-based DC

Millimeter-Wave (mm-Wave) UAV communications suffer from many issues, such as blockage in ground environments, highly dynamic aerial and ground networks, movements of obstacles, reflection, and the orientations of devices. It has also been distinguished that the resource allocation in the lower layers is not considered and, even worse, in most cases, not coupled. Furthermore, the performance metrics for mmWave-UAV-based DC are not defined beforehand and are not optimized as they should be. As a possible solution, Beamforming

Table 18 (continued).

	Problems	Proposed solutions	Recommended references
Blockchain-based security for UAV-based DC	• UAVs cannot provide the required computation power for blockchain functioning. • Blockchain-based UAV applications are more vulnerable to more-advanced types of attacks. • It is very complicated to implement blockchain-based UAV applications due to the interoperability problem of UAV networks. • The quality of wireless links between UAVs could have a crucial impact on the blockchain structure.	Potential solutions could be studied to increase the computation efficiency of UAVs using, e.g., a Fog architecture while guaranteeing safer blockchain structures. In addition, ultra-reliable transmission models could be proposed to ensure efficient interoperability within UAV networks. A significant effort should be dedicated to making an in-depth analysis of the proposed solutions.	Mehta et al. (2020), Alladi et al. (2020), Alkadi et al. (2022), Chang et al. (2021), Aggarwal et al. (2020)
Satellite-UAV-based DC	• Satellite-UAV communications cause more significant time-varying De. • Satellite-UAV communications are characterized by their fading channels. • Due to their long transmission latency, it is very challenging to deploy real-time applications involving satellite-UAV communications. • The high mobility of UAVs negatively impacts multi-access asynchronous transmissions in satellite-UAV communications.	AI techniques could be adopted to mitigate fading problems and address most physical layer operations. Moreover, advanced time-varying channel estimation should be carefully designed to overcome time-varying (De) and get accurate CSI and beamforming angles. A deep investigation is required to address the different issues of high-mobility satellite-UAV communications.	Lee et al. (2022), Tian et al. (2021), Liu et al. (2018b), Tang et al. (2022), Gu et al. (2020), Li et al. (2020a), Xiao et al. (2022), Kong et al. (2022), Guo and Zhao (2022), Fang et al. (2022), Zhou et al. (2022b), Ma et al. (2021), Jia et al. (2020)

technology is perfectly appropriate for mmWave-UAV communications where antenna arrays operate in high-frequency bands. This technology allows the communicating nodes to perform flexible beamforming while compensating for the high propagation loss of target signals and mitigating interferences at the level of UAVs.

7.4. Interference in massive UAV-assisted networks and IoT

The enormous expansion of end devices and their different requirements in data throughput, freedom of mobility, and spectral efficiency will lead to the deployment of many UAVs to optimally serve the network. However, a high density of drones will directly lead to increased interference, which may significantly impact the network's performance and transmission capacity. Moreover, all UAVs within a given network use the shared wireless channel for exchanging information with the different entities of the network, causing complex wireless interferences that are sometimes impossible to avoid. From another perspective, two critical sources of interferences are also distinguished: (i) interference between UAVs and terrestrial macro cells and (ii) interferences between solely UAVs, where the coverage of one node overlaps another. Decreasing the level of interference could significantly improve the network throughput and energy consumption. For this purpose, new learning techniques need to be designed to have a great degree of autonomy, less information sharing, and self-organized capacity. Furthermore, distributed schemes are also helpful in reducing the computational overhead, thus less information exchange.

As for the IoT devices, it may also lead to the problem of interference especially when they are deployed in large-scale in the context of smart cities and wearable (Peng et al., 2019). Moreover, since IoT devices could be served by UAVs during the DC process, the interference problem should be mitigated using the appropriate techniques, such as in Jee and Prakriya (2022) where the authors proposed an excellent method for interference mitigation based on NOMA.

7.5. Resource and spectrum scheduling for UAV-based DC

Spectrum management and UAV resources should be elaborately scheduled to enhance the quality of UAV communication and services. Indeed, the spectrum efficiency strongly depends on the altitude of UAVs, their 3D coverage, and their energy efficiency, which are

not always considered in the design of schemes. In addition, various elements of resource management, three of which have to be considered. First, backhauling through UAVs can be cost-efficient since no infrastructure is required, but it suffers from many constraints, such as UAV placement, security, interferences, and communication with cellular BSs. The second is offloading through UAVs, where data can temporarily be transferred to UAVs at peak times. This increases the system's performance to efficiently serve the increased demand of users. However, the most distinguished challenges in data offloading are relay energy and delay, which are required to ensure a specific data transfer rate and target allocations. Finally, the placement of UAVs is a crucial element of resource management, directly impacting the system's transmit power, QoS, and coverage. Nevertheless, the energy consumption of UAVs, mobility of ground devices, obstacles (e.g., buildings), and environmental restrictions all constitute a severe challenge to overcome. Possible solutions to these issues could adopt heuristic algorithms, AI methods, or even existing optimization algorithms while considering the appropriate metrics to jointly optimize UAVs' trajectories, placements, energy consumption, and transmit power.

7.6. Blockchain-based security for UAV-based DC

Blockchains are robust against a variety of security threats against UAV networks. However, it is well-known that UAVs have restricted computation energy while blockchain requires high computation energy to execute, e.g., consensus algorithms on UAVs. It is also distinguished that most blockchain-based UAV applications need permissioned blockchains, which are more susceptible to attacks than public blockchains with many more participants. Moreover, such blockchain-based UAV applications are challenging to be implemented due to the lack of an interoperable solution for UAV networks. To top it off, blockchain-based UAV applications are also impacted by access control and radio link quality, which can strain the consensus process and the information flow between nodes. To overcome these challenges, it is required to find solutions to increase the computation efficiency of UAVs while making blockchain structures safer and immutable. Furthermore, trusted UAV relays could be considered a promising option while ensuring efficient and ultra-reliable transmission models to achieve interoperability within UAV networks using blockchain. All these solutions are worthy of serious and sustained effort from researchers.

7.7. Satellite-UAV-based DC

Most UAV applications cannot be deployed without terrestrial networks which, in turn, cannot be implemented anywhere (Wang et al., 2022d). This constraint creates a bottleneck for further UAV development and use. Fortunately, satellite networks can ensure large backhaul coverage for UAVs, which has the potential to spur the development of more novel applications. For instance, this kind of architecture can support many applications, such as forest fire recovery, mineral exploration, post-disaster management, and other applications. However, it is clearly observed that the need for satellite-UAV communications are increasing as the range of UAV applications expands. Also, the unique features of satellite-UAV communications, such as large Doppler spreads, long propagation latency, and deep fading, make research on such kind of architecture very challenging. To address these challenges, multi-user access schemes, advanced anti-Doppler transceivers, and accurate channel estimation, should all be considered. In addition, AI techniques could be adopted to optimize transceivers functioning and multi-access systems. These solutions are considered as potential research directions in the future.

8. Conclusion

The UAV-based DC concept has prompted a keen interest in achieving monitoring, coverage, data offloading, and relay of information. In this paper, we examined the opportunity presented by the UAV-based DC concept to be integrated into different UAV-assisted network applications through a comprehensive survey and detailed discussions based on the growing research topics in the field. The drive behind this study is the lack of detailed and complete surveys on UAV-based DC. To bridge this gap, we first introduced a background of the UAV-based DC concept and its elements to allow readers to comprehend better how the DC process through UAVs is performed. We then provided a comprehensive analysis of recent UAV-based DC schemes classified according to our proposed taxonomy, along with their objectives, advantages, and drawbacks. From this analysis, several key lessons learned have been summarized, and the evaluation tools used have been highlighted. Finally, in an attempt to integrate the concept of UAV-based DC into recent hot topics, interesting issues, future directions, and recommended references have been outlined. We believe this work will help researchers learn more about the cutting-edge topic of the UAV-based DC concept and promote more research efforts towards designing efficient UAV-based DC schemes in the discussed research directions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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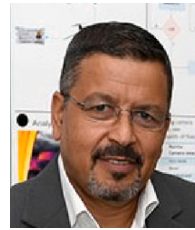
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